GOLDEN FORESTS GROUP

BUSINESS PLAN

Premium Agroforestry Investment Platform and Plantation Management

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Registered Entity: Golden Forests Agroforestry Intelligence Investment in Agricultural Enterprises & Management FZCO Registered in Dubai, United Arab Emirates.

Safe Harbour Statement.

The information contained in this document may contain and include forward-looking statements related to our future growth, trends in our industry, our financial and or operational results, and our financial or operational performance. Such forward-looking statements are predictive in nature, and may be based on current expectations, forecasts or assumptions involving risks and uncertainties that could cause actual outcomes and results to differ materially from the forward-looking statements themselves. Such forward-looking statements may, without limitation, be preceded by, followed by, or include words such as "believes", "expects", “anticipates", “estimates”, "intends", “plans”, or similar expressions, or may employ such future or conditional verbs as "may", "will", "should" or "would", or may otherwise be indicated as forward-looking statements by grammatical construction, phrasing or context. The risks and uncertainties referred to above include, but are not limited to: variability of operating results among periods; inability to retain or grow our business due to execution problems resulting from significant headcount reductions, plant closures and product transfer associated with major restructuring activities; the effects of price competition and other business and competitive factors generally affecting the timber and power generation industry; the challenges of effectively managing our operations during uncertain economic conditions; our dependence on a limited number of customers; our dependence on industries affected by rapid technological change; the challenge of responding to lower-than-expected customer demand; our ability to successfully manage our international operations; and delays in the delivery and/or general availability of various components used in the manufacturing process. Except as required by applicable law, Golden Forests disclaims any intention or obligation to update or revise any forward-looking statements, whether because of new information, future events or otherwise.

ABOUT THIS DOCUMENT

Company Name and Structure

Golden Forests is the trading name of Golden Forests Agroforestry Intelligence Investment in Agricultural Enterprises & Management FZCO, registered in Dubai, United Arab Emirates. Throughout this business plan, "Golden Forests," "the Company," and "the Group" refer collectively to the Golden Forests group of companies, including the ADGM holding company (Abu Dhabi), the Dubai-based sales and marketing entity, and Crassna Agroforestry Development Incorporated (CADI), the plantation management company in the Philippines.

Key Terminology

Term	Description
Agroforestry intelligence	Golden Forests' proprietary operational framework combining agronomic science and AI-enabled precision agriculture.
Agarwood	The resinous heartwood produced by Aquilaria trees, highly valued for its fragrance and use in luxury products.
Aquilaria crassna	The primary species of agarwood cultivated by Golden Forests, selected for its resin quality and market preference and approved by the Bureau of Customs for Importation and Export.
Carabao Mango	A premium mango variety from the Philippines, known for its sweetness and export value.
Clients	Purchasers of individual agarwood or mango trees who receive ownership of specifically identified biological assets managed by the Golden Forests ("The Company").
CITES	Convention on International Trade in Endangered Species, which regulates the trade of endangered plants and animals, including agarwood.
Copyrights	The option to market and sell the proprietary inoculation formulation under private label brand to other potential clients within the Philippines.
Commission revenue	The percentage of sales revenue retained by the company for managing the harvest and sale of products on behalf of tree owners.
Durian	A large, tropical fruit distinguished by a thick, sharp, thorn-covered husk and a notoriously intense aroma.
Dwarf Sweet Elena Carabao Mango	A new premium variety of mango produced from the grafting of Carabao Mango seedlings with Elena bud stock and applying dwarfing techniques.
Forestry Industry Organization (FIO)	A state enterprise in Thailand, established to manage natural resources, economic forest plantations, and wood industries sustainably.
Harvesting	Harvesting of agarwood trees refers to cutting down the tree which occurs in either year 9 or 10 from the date of out planting. Harvesting of mango trees refers to the picking and collection of mango fruits.
Hectare	One hectare represents 10,000 square meters and the equivalent of 2.47 acres
High-density planting	An agricultural practice where more trees are planted per hectare than traditional methods, increasing productivity.
Investors	Equity or debt investors in Golden Forests ("the Company").
Inoculation	The process of inducing resin formation in agarwood trees, typically by introducing a specific type of infection or stress.
Intellectual property	The exclusive sublicensing rights to use inoculation formulations and other related technologies for *Aquilaria crassna* agarwood plantations in the Philippines.

Induced flowering	Techniques used to stimulate mango trees to flower and fruit continuously, year-round, bypassing their normal season cycle.
Marketing	A multi-touch attribution model to measure marketing effectiveness across digital, content, agent, event, and organic channels, enabling precise optimisation of client acquisition spend while building long-term brand equity in the Golden Forests name and registered trademark products.
Mortality buffer	Surplus trees planted to replace losses occurring from the natural causes identified in this document, thereby safeguarding clients' holdings.
Wood	Another term for agarwood, especially when referring to its use in incense and perfumes.
Oud wood chips	Highly aromatic, resin-soaked pieces of wood, extracted from inoculated *Aquilaria crassna* agarwood trees.
PEFC/FSC accreditation	International certifications for sustainable forest management (Programme for the Endorsement of Forest Certification / Forest Stewardship Council).
Reforestation programme	The Company's initiative to plant native tree species for every commercial tree sold, supporting ecosystem restoration.
SPV (special purpose vehicle)	A subsidiary created for a specific investment purpose, often used to isolate financial risk.
Tokenised investment vehicles	Future capital structures that may include digital tokens representing ownership or investment, subject to regulatory approval.
Yield	The amount of product (e.g., oud wood chips, mango fruit) produced per tree.

Appendices

Supplementary sections in the business plan providing detailed financial forecasts, technical data, and supporting information.

Currency

All financial figures are expressed in United States Dollars (USD) unless otherwise stated.

Measurements are expressed in metric units (kg, hectares, mm).

Nomenclature

Scientific names are presented in accordance with international nomenclature conventions.

Language

All text is written using UK spelling. British English grammar and formal business style are applied throughout. Dates follow the DD Month YYYY format, and units of measurement use the International System of Units (SI) unless otherwise stated. Acronyms are defined upon first use and applied consistently.

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EXECUTIVE SUMMARY

Strategic Value of Agarwood and Carabao Mango in Sustainable Agroforestry

Aquilaria crassna (agarwood) and *Mangifera indica* (Carabao Mango) are high-value assets within sustainable agroforestry, supported by established global demand across luxury fragrance and premium food markets. Their cultivation offers differentiated income potential, combining the long-term value appreciation of rare, aromatic luxury commodities with the recurring revenue profile of highly sought-after, nutritious fruit products.

Both agarwood and Carabao Mango possess deep cultural and commercial significance across Southern and Southeast Asia, as well as key Middle Eastern markets. This sustained demand profile reinforces long-term market depth across luxury fragrance, wellness, and premium food supply chains. The integration of precision farming technologies has revolutionised the establishment and management of private plantations, making it feasible for both private and professional clients to participate. These advancements support the efficient cultivation, monitoring, and harvesting of these trees, thereby enhancing returns across the entire life cycle of plantation investments.

The Philippines as a Strategic Production Hub of Agarwood and Mango

The Philippines is strategically positioned to emerge as a leading supplier of agarwood (oud) wood chips and premium mango fruits. Its tropical climate, favourable rainfall patterns, and diverse soil conditions create optimal environments for the cultivation of both species, supporting consistent quality and yield performance. In addition, government incentives supporting agricultural development contribute to sector growth, encouraging both local and international capital.

An experienced agricultural labour force supports operational efficiency, cost control, and consistent quality standards across plantation management. Production frameworks are structured to meet international grading and export requirements, enabling competitive positioning alongside established global suppliers. Leveraging its natural advantages and regulatory framework, the Philippines maintains a strong position within the international agarwood and mango supply chain.

Intellectual Property

Golden Forests owns the exclusive sub-licensing rights to the use in the Philippines of the inoculation formulations and other technologies designed by the Forestry Industry Organization (FIO) of Thailand for *Aquilaria crassna* agarwood plantations, and distributed by Ecoyield Service Co. LTD, a company registered in Thailand. The Company retains the rights to acquire the copyrights and be entitled to market and sell the inoculation formulation under its own private label brand to other potential clients within the Philippines.

The Company was the first Company to be granted the permitting rights to introduce *Aquilaria crassna* agarwood (an exotic species) into the Philippines by the Secretary of the Department of Environment and Natural Resources (DENR), and approval to Import and Export by the Bureau of Customs (BOC) in the Philippines. The Company intends to cultivate *Aquilaria crassna* under a registered trademark name.

Golden forests have developed a proprietary strain of mango, produced from the grafting of Elena bud stock with Carabao seedlings and applying dwarfing techniques which allow for planting at high density. The new variety is referred to as Dwarf Sweet Elena Carabao Mango. The Company intends to create a brand name for this variety and trademark the brand name.

Plantation Management and Investment Opportunities

Our plantation specialists bring more than 80 years of combined experience in commercial plantation development, cultivation, and operational delivery. Plantation strategy and species-specific methodologies are guided by leading scientists from prominent Asian universities, whose expertise supports advanced cultivation practices and inoculation protocols, as described in section 8 (*Company Organisation*). These scientific inputs are integrated into our AI-driven plantation management systems, enhancing precision, monitoring, and performance across the cultivation lifecycle.

We offer professionally managed plantation investment opportunities for both private and professional clients. Future capital structures may include tokenised investment vehicles and bond- or SPV-based structures, subject to regulatory approval.

Agarwood plantations are expected to achieve around 18% annual returns over a 10-year period (*see Appendix B*). Each tree is projected to produce about 1.5 kg of oud wood chips when harvested in year nine and 2.0 kg when harvested in year ten (*see Appendix C*), based on standard cultivation methods used in the Company's financial models. Annual returns reflect 50% of a client's tree portfolio harvested in year 9, and the balance in year 10.

Mango plantations typically last 25 years, with estimated returns of 12% for the first 15 years and over 20% during the final 10 years (*see Appendix B*). Trees start producing fruit by year five, yielding about 30 kg each, increasing to a stable 100 kg annually from year ten onwards (*see Appendix D*). Induced flowering enables year-round harvesting, reducing seasonal risks and ensuring steady market supply.

Scalability and Diversification Growth Initiatives

Golden Forests has signed a Letter of Intent with the FIO, represented by its agents, Ecofield Service Co., LTD, to establish privately managed *Aquilaria crassna* agarwood plantations on land owned by the FIO and located within 100km of Bangkok, Thailand. The FIO will be responsible for providing the land, supply of seedlings, out-planting and maintenance of the trees over the 10-year life cycle. The Company will pay a fixed fee per tree per annum and be responsible for all costs associated with inoculation, harvesting and extraction of the oud wood chips.

The Company intends to diversify and commence the cultivation of durian fruit trees in 2027. The variety of durian in the Philippines is noted for its aroma, which lacks the strong odour typically associated with the fruit, an odour that has led to bans on public transport and restrictions by several international freight carriers. A detailed assessment of durian for future cultivation and diversification (*see Appendix I)* has been completed, along with an overview of the cultivation in the Philippines (*see Appendix J*).

Project Timeline and Agroforestry Innovation

Agarwood seedlings will be planted in July 2026, following a dedicated period of nursery propagation. The company has leased land in southern Zambales and possesses exclusive licensing rights to advanced agarwood

inoculation formulas and techniques tailored specifically to *Aquilaria crassna*. These methods, developed in collaboration with scientists from the FIO, have demonstrated efficacy in enhancing both the yield and quality of oud resin. This ensures consistent production of premium oud wood chips while supporting sustainable practices in agarwood cultivation. Each hectare of plantation will accommodate 1667 agarwood trees (*3 m x 2 m spacing*). For the initial two years post-planting, cassava will be intercropped to provide shade for the developing agarwood trees. After the initial two years, the cassava will be harvested and donated to the local communities where our plantation workers originate from.

A pilot Carabao Mango plantation was launched on Panay Island in September 2025. Through proprietary grafting and dwarf techniques, a new dwarf mango variety combining Carabao and Elena genetics has been developed, with initial planting scheduled for July 2026. This premium variety is referred to as Dwarf Sweet Elena Carabao Mango. Mango fruits are typically harvested annually. However, through the implementation of flower induction techniques, year-round flowering, and therefore harvesting can be achieved, which mitigates seasonal volatility and supports more stable revenue generation. The Company plans to expand its mango propagation programme and implement advanced monitoring systems to enable real-time management of soil conditions, irrigation and pest control. This strategy is designed to enhance productivity, crop quality and sustainability while reinforcing operational excellence. Each hectare of mango plantations will accommodate 416 trees (*6 m x 4 m spacing*).

Financial Highlights

The financial model for Golden Forests, shown in Appendix A, outlines a fifteen-year projection covering agarwood and mango cultivation, alongside native species reforestation. Projected cumulative gross sales total approximately USD 1.3 billion, with net sales of USD 574.9 million after deductions. Revenue growth is driven by increasing productive tree volumes and projected market pricing assumptions. Export sales of oud wood chips commence in year nine, while mango fruit sales begin in year five, with approximately 70% of mango production allocated to export markets.

Total cost of sales over the period is projected at USD 125.4 million and operating expenses at USD 20.3 million, with direct materials and labour representing the primary expense categories. Gross profit margin is projected at 44.2%, with a net income margin at 30.5%, resulting in cumulative net income after interest and taxes of USD 321.8 million. Net margin is influenced by the capitalisation of certain pre-operational agarwood cultivation costs, which are recognised after a seven-year establishment period in accordance with the Company’s accounting assumptions.

The balance sheet reveals a robust financial position, with total assets reaching USD 336.4 million, the vast majority in current and highly liquid forms such as cash and equivalents. Liabilities are minimal at USD 14.4 million, all of which are current, leaving the company free of long-term debt at year-end. Equity stands strong at USD 322.0 million, primarily fuelled by substantial retained earnings, underscoring the company's steady profitability and prudent financial management throughout the projection period.

The financial model incorporates sustainability measures, including reforestation allocations, mortality provisions, and structured investor return mechanisms. Strategic investment in irrigation, security, and agricultural technology supports operational efficiency. Tax and VAT incentives are not included in the projections and are currently subject to regulatory approval.

Reforestation Programme

Golden Forests has incorporated a structured reforestation programme focused on denuded areas in Southern Zambales. The programme supports ecosystem restoration and biodiversity enhancement through the planting of endemic species, including Narra, Agoho, Mangono, Molave, Talisay and Belite.

For every agarwood and mango tree acquired by clients, the Company allocated the planting of one endemic tree species. This one-to-one reforestation framework ensures that commercial plantation development is directly linked to native forest regeneration within the region.

Client Plantation Visit Programme

Golden Forests offers a structured client plantation visit programme, enabling clients to visit operational sites in Southern Zambales, Philippines. The plantations are located approximately one and a half hours from Clark International Airport, providing efficient access for both domestic and international visitors.

The Company hosts a two-day site programme, including guided plantation tours, operational briefings, and discussions with management and technical teams. This initiative supports transparency, operational oversight, and informed participation in plantation investments.

All travel and accommodation arrangements, including those required to reach or depart from the hosted programme, remain the sole responsibility of the client.

1. COMPANY OVERVIEW

Business Structure and Market Focus

Golden Forests operates as an agroforestry investment and plantation management platform. The Company is currently focused on premium agarwood and mango cultivation, with a scalable model designed to incorporate additional high-value agroforestry species as part of a disciplined growth strategy. The business integrates plantation development, operational management, and structured investment participation designed for private and professional clients. This platform approach supports efficient capital deployment, optimised land utilisation, and long-term biological asset development.

Global Corporate Structure

Golden Forests operates through an ADGM-based holding company established in Abu Dhabi, United Arab Emirates. The Company's head office is located at Level 25, Al Sila Tower, Al Maryah Island, ADGM, Abu Dhabi, UAE.

The ADGM holding company provides Group-level governance, capital structuring, and strategic oversight. It coordinates the activities of the Group's operating and functional entities, including the Dubai-based international sales and marketing company and the Philippine plantation operating company. This structure establishes clear

separation between (i) Group governance and funding oversight, (ii) international investor engagement and market development, and (iii) operational and plantation management in the Philippines.

A key responsibility of the ADGM holding company is oversight of funding designated for agarwood inoculation, harvesting, and extraction activities. Such funds are maintained within a structured, interest-bearing framework and released in accordance with predefined lifecycle and operational milestones, supporting financial discipline and transparency throughout plantation development.

International Sales and Marketing Operations

Golden Forests Agroforestry Intelligence Investment in Agricultural Enterprises & Management FZCO has been established in Dubai, United Arab Emirates. The registered office of the Dubai entity is located at IDZA Business Park, DDP 75773-001, A1-3641379065, Dubai Digital Park, Dubai Silicon Oasis, Dubai, United Arab Emirates. This entity serves as the Group's international sales and marketing company and operates in alignment with the governance framework established by the ADGM holding company.

The Dubai entity is responsible for international client engagement, relationship management, and market development across priority regions. Sales agents located in the Middle East, Europe, the United Kingdom, and Southeast Asia support regional outreach, while a structured digital communications platform facilitates client onboarding and ongoing engagement.

This structure enables coordinated client engagement while maintaining clear separation between Group governance, market development, and plantation operations in the Philippines.

Plantation Management

Crassna Agroforestry Development Incorporated (CADI) serves as the Group's plantation operations company, responsible for the implementation and management of all plantation activities. Its registered head office is located at Level 24, Philippines Stock Exchange Tower, One Bonafacio Street, 5th Ave Cor.28th Street, BGC, Taguig City, Philippines.

CADI oversees the full lifecycle of plantation operations, including land preparation, nursery coordination, planting, intercropping, inoculation, maintenance, harvesting, and local logistics. Operational activities are conducted in accordance with the Group's agronomic standards, sustainability framework, and quality control protocols.

In addition to plantation execution, CADI coordinates local supply chain management and export preparation, ensuring that agarwood and mango production align with international market requirements. Reporting and operational alignment are maintained with the ADGM holding company and the Dubai-based sales entity to support governance transparency and structured client oversight.

Vision

To lead the future of sustainable agroforestry in Asia through Agroforestry Intelligence, setting the benchmark for premium, high-value crops by uniting artificial intelligence, advanced science, and best-in-class genetics to create long-term value, resilient ecosystems, and thriving local communities.

Mission

Our mission is to position sustainable agroforestry as a compelling alternative investment opportunity, delivering outstanding long-term returns while creating meaningful local impact and environmental sustainability.

We achieve this by establishing a thriving and ethical agroforestry enterprise that transforms Asian tree plantations into internationally recognised, high-value assets, harnessing the unique potential of premium agroforestry crops. Through sustainable practices, world-class production standards, and community partnership, we generate enduring value for clients, ecosystems, and the communities we serve.

Objectives

- Generate disciplined, long-term returns through the application of advanced agroforestry science and AI-enabled plantation management systems.
- Maintain full regulatory compliance across all operating jurisdictions, establishing a robust governance framework for cross-border agricultural investment.
- Build a geographically diversified client base through disciplined regional expansion and strategically aligned institutional and strategic partners.
- Develop scalable plantation models that integrate sustainability, yield optimisation, and responsible land stewardship.
- Uphold rigorous governance, financial controls, and transparent reporting standards to support long-term client confidence.

Core Values

Golden Forests operates in accordance with a clearly defined set of values that underpin its governance framework, operational standards, and long-term strategic direction.

Integrity

We uphold transparency, accountability, and ethical conduct across all jurisdictions in which we operate.

Operational Discipline

We apply structured management systems and rigorous oversight to ensure consistency, efficiency, and performance across plantation development and client engagement.

Responsible Stewardship

We are committed to sustainable land use, environmental protection, and full compliance with applicable regulatory standards, ensuring responsible cultivation practices at every stage of the plantation lifecycle.

Quality and Precision

We employ scientifically grounded propagation and cultivation methodologies to deliver consistent, high-grade agricultural output aligned with international market requirements.

Long-Term Value Creation

We take a long-term view, aligning sustainable plantation performance with responsible investment management.

Community Development

We engage local communities through employment, skills development, and social initiatives that contribute to long-term regional resilience.

Agroforestry Intelligence and Operational Technology:

Golden Forests operates through its proprietary Agroforestry Intelligence framework, which integrates agronomic science, structured plantation management, and AI-enabled monitoring systems to optimise biological asset performance across the cultivation lifecycle.

The framework employs data analytics, remote sensing, machine learning, and predictive modelling to support evidence-based decision-making. AI-enabled systems monitor soil conditions, climate patterns, pest activity, and growth progression, enabling precise irrigation scheduling, targeted nutrient application, and optimised harvest timing.

Agricultural drones provide aerial surveys for large-scale crop health assessment, stress pattern identification, and yield estimation. Data collected through drone operations is integrated into the Agroforestry Intelligence platform, strengthening operational visibility and risk management.

This technology-enabled approach supports yield predictability, resource efficiency, and environmental stewardship whilst reducing input costs and operational risk. Detailed technical specifications and operational applications are outlined in Appendix H.

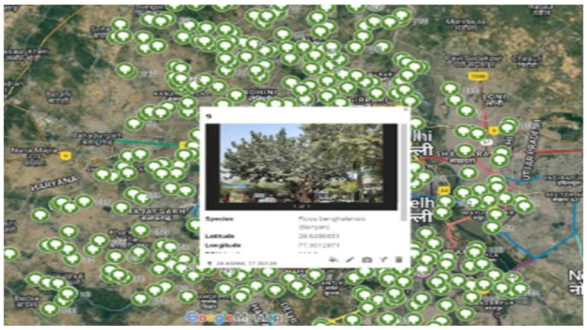
Agricultural Drones

Crop monitoring, field mapping, fertiliser / Pesticide spraying and plant health assessment

Smart Soil Sensors

Smart Irrigation Systems

Satellite Geotagging



Inoculation Formulae’s

2. INDUSTRY AND MARKET ANALYSIS

Comprehensive Overview of the Agarwood Industry

The global agarwood industry is demonstrating consistent expansion, with current market valuations exceeding USD 41 billion (*Persistence Market Research 2024*). Projections indicate the market will reach approximately USD 69 billion by 2029 and USD 84.9 billion by 2034 (*Precedence Research 2024*).

Growth is driven by rising demand across luxury fragrance, religious, medicinal, and speciality product markets, with concentration in the Middle East, South Asia and East Asia.

Agarwood derives its commercial value from the resin formed within Aquilaria trees following inoculation or natural infection. As the resin develops and infuses the heartwood, the material can be harvested and processed into oud wood chips, traditionally burned as incense or distilled into essential oils. These materials rank among the most highly valued aromatic substances globally.

The Philippines offers favourable conditions for *Aquilaria crassna* cultivation, including a tropical climate with appropriate rainfall distribution (1,500–2,500mm annually), well-drained soils, and competitive agricultural labour costs. Government support for agricultural development, including tax incentives and streamlined export facilitation for CITES-compliant agarwood, further supports operational viability. However, Golden Forests' cultivation methodologies and Agroforestry Intelligence framework are designed to be geographically adaptable, with potential application across other Southeast Asian and South Asian regions exhibiting suitable agro-climatic conditions.

Commercial applications include oud wood chips, distilled essential oils, incense, carvings, and high-end speciality items. Premium-grade oud oil commands the highest pricing within the value chain, with top-quality agarwood essential oil reported to reach prices of up to USD 40,000 per litre (*ScienceDirect 2020-2021*), reflecting premiums placed on purity, resin concentration, and authenticated origin.

The industry remains supply-constrained due to historical overharvesting of wild Aquilaria species and increasing regulatory oversight under international trade frameworks, including CITES. Consequently, plantation-cultivated, traceable agarwood has become the preferred source of supply in global markets, particularly for buyers requiring documented sustainability credentials and export compliance.

Demand is underpinned by deep-rooted cultural and religious traditions across the Middle East, South Asia, and East Asia, alongside growing international interest in natural fragrances, luxury goods, and holistic wellness products.

Target Export Markets

Global agarwood trade concentrates in specific regional markets, each demonstrating distinct purchasing behaviours and end-use applications:

Middle East (United Arab Emirates (UAE), Saudi Arabia): The Middle East represents the largest and most established market for agarwood products. The UAE and Saudi Arabia are major importers of oud wood chips and

distilled essential oils, driven by sustained demand within religious, cultural, and luxury fragrance sectors. Agarwood is widely used in traditional incense (bakhoor) and high-end perfumery, supporting consistent premium pricing for high-grade material.

East Asia (China, Taiwan, Japan, Singapore): East Asian markets demonstrate diversified demand across medicinal, religious, and luxury applications. China maintains traditional uses in medicine and spiritual practices, reinforcing stable baseline demand for both raw chips and refined oils. Taiwan and Singapore function as significant processing, trading, and redistribution hubs within the regional supply chain. Japan represents a premium market for authenticated high-grade oud products.

Europe (United Kingdom (UK), Germany): European demand remains more niche but expanding, particularly within luxury fragrance, alternative wellness, and speciality incense markets. The UK and Germany are leading European importers of pure oud oil, supported by growing consumer interest in natural and high-value aromatic products.

United States (US): The US market is emerging as a growth region for agarwood products, driven by increasing consumer interest in natural fragrances, incense, and holistic wellness applications. While smaller than Middle Eastern markets, the US presents long-term diversification potential for sustainably sourced, plantation-grown agarwood.

Competitive Analysis

The Philippines has strong potential as a supplier of oud wood chips to premium international markets. Success within this environment depends on meeting strict import and phytosanitary standards, ensuring certification and traceability, and building efficient logistics and trusted partnerships. Adaptability and strong buyer relationships are vital.

Product differentiation through unique agarwood varieties and sustainable practices can help capture niche markets and justify higher pricing tiers, particularly when supported by quality-controlled processing and value-added products. Developing a recognised “Philippine Oud” brand supported by origin labelling and sustainability credentials, and leveraging the Philippines unique biodiversity, has the potential to strengthen global buyer confidence and market positioning.

Prospects of growth exist in emerging markets like the US and Europe, while established demand remains concentrated in the Middle East and across Asia. However, sustained competitiveness will require continued adherence to evolving regulatory standards, responsiveness to rising sustainability expectations, and disciplined investment in supply chain infrastructure. Maintaining compliance, strengthening buyer relationships, and supporting continuous operational improvement will be essential to securing long-term positioning within premium export markets.

Aquilaria crassna: Strategic Species Selection and Competitive Positioning

Golden Forests has selected *Aquilaria crassna*, as the primary species for cultivation, reflecting market preference and the performance characteristics required for premium-grade oud production. This selection supports the

Company's focus on consistent resin formation, quality control, and alignment with premium export market requirements.

Golden Forests sources *Aquilaria crassna* genetic stock from Thailand and utilises DNA-accredited crassna seedlings. This approach assures species authenticity and strengthens traceability across the supply chain. This ensures a reliable foundation for the application of proven inoculation methodologies, which are central to inducing resin production and achieving consistent grading outcomes.

By combining verified genetics with validated cultivation and inoculation techniques, Golden Forests is positioned to produce high-quality oud wood chips that meet the stringent standards expected in global markets.

A detailed comparison of the characteristics and competitive advantages of *Aquilaria crassna* and *Aquilaria malaccensis* is provided in Appendix G.

Competitive Landscape

The agarwood industry operates within a competitive environment shaped by regulatory compliance, capital requirements, and technical expertise in resin induction and grading. Long cultivation cycles and export certification standards contribute to increasing professionalisation within the sector.

Product differentiation: Agarwood pricing is driven by resin quality, grading consistency, and aromatic profile. Premium-grade oils command the highest pricing tiers and are supplied to established buyers in the Middle East, Europe, and high-end Asian markets.

Supply chain complexity: Most agarwood exports are traded as wood chips and processed into oils in destination markets such as the Middle East, Singapore, and Taiwan. Reliable logistics, traceability, and long-term trading relationships are critical to maintaining market access.

Regulatory and sustainability pressures: Agarwood is regulated under international trade frameworks due to overharvesting and conservation concerns. Producers that demonstrate compliance, certification, and documented cultivation practices are better positioned to secure stable export channels and gain a competitive edge.

Market access and buyer relationships: Success depends on robust buyer partnerships and the ability to meet strict import and certification requirements, particularly in markets with established religious, medicinal, or luxury uses for agarwood products.

Global Mango Industry Trends and Market Outlook

The global mango market size stands at USD 55.20 billion in 2025 and is projected to reach USD 68.30 billion by 2030 (*Moore Intelligence 2025*), reflecting a 4.4% CAGR through the forecast window. The market benefits from rising demand in regions like North America and Europe, supported by improved logistics and refrigeration. Consumer preferences for healthy, tropical fruits are boosting mango popularity, alongside culinary versatility and health benefits. Importers and exporters can capitalise on these trends through quality control measures, market diversification, and digital platforms to meet global demand effectively. Wider supermarket penetration and steady urban income growth are also propelling the fruit from regional staple to global commodity.

Growth projections vary because of the rising popularity and demand for mangoes globally each year. Tridge, a leading global agri-food data, intelligence and sourcing platform that connects buyers and suppliers across more than 150 countries, projects the global market size to see strong gains by a CAGR of 7.1% to USD 89.34 billion by 2028 (*Tridge 2025*).

Technology that extends shelf life, such as controlled-atmosphere containers, now allows distant suppliers in South America and Southeast Asia to compete in premium import markets. At the same time, digital produce platforms are compressing supply-chain layers, enabling small farms to capture more value and meet retailer traceability rules.

The collaboration among stakeholders across the mango value chain, including large-scale producers, exporters, regional aggregators, nursery developers, and post-harvest solution providers, contributes to market development. Export-focused regions are improving orchard management, developing new varieties, and enhancing quality control while investing in cold-chain logistics, certification systems, and value-added processing.

Comprehensive Analysis of the Philippine Mango Industry

The Philippine mango industry maintains a well-established position within the global fruit export sector, supported by competitive pricing, exceptional quality standards, distinctive varietal characteristics, and recognised origin branding. The Carabao Mango variety is internationally recognised for its flavour profile and export appeal.

Philippine mango export performance has demonstrated sustained growth over the past two decades. In the early 2000s, annual export values were estimated at approximately USD 35 million (*Food and Agriculture Organization of the United Nations 2004*). By the mid-2010s, annual export revenues had surpassed USD 50 million (*American International Journal of Business Management 2020*), supported by sustained demand in the US, Japan, South Korea, and the Middle East. As of the mid-2020s, the export revenues are estimated in the range of USD 1.6 billion annually (*Mordor Intelligence 2025, Research and Markets 2025, Worldmetrics 2025*).

Philippine Industry Overview

The Philippines ranks among the world's leading mango producers and exporters, supported by favourable agro-climatic conditions and established cultivation practices. Production concentrates in key regions including Guimaras, Zambales and Pangasinan, where growers focus on delivering consistent quality standards and compliance with international export requirements.

Export volumes are influenced by seasonal harvests, weather variability and ongoing improvements in post-harvest handling and logistics infrastructure. Philippine mangoes are marketed for their sweetness, vibrant colour, and firm texture, characteristics that underpin their premium positioning in international markets.

Market Positioning and Value Drivers

The export value of Philippine mangoes is influenced by competitive pricing, recognised origin branding, and consistent product quality. Premium pricing is supported by disciplined grading standards and the ability to meet stringent import and phytosanitary requirements in destination markets.

Export pricing is shaped by global demand dynamics, production costs (including labour, logistics, and regulatory compliance), and supply reliability across harvest cycles. Philippine mango exporters maintain value through quality consistency, export discipline, and responsiveness to evolving consumer and regulatory expectations.

Target Export Markets

Philippine mango exports are concentrated in several high-value destination markets:

United States: A primary destination market, supported by steady demand for premium fresh fruit and well-established import and distribution channels. The US market values consistent sizing, appearance standards, and reliable supply windows.

Japan and South Korea: These markets place strong emphasis on food safety, traceability, and presentation standards. Exporters meeting rigorous certifications and grading requirements access premium pricing tiers, with Japanese markets particularly sensitive to blemish-free fruit and premium packaging.

Middle East: Demand is supported by consumer preference for sweet mango varieties and established trade relationships, particularly during peak harvest periods. Middle Eastern markets demonstrate less price sensitivity for premium-grade Carabao Mangoes compared to competing varieties.

Across these markets, purchasing decisions are influenced by grading consistency, visual quality, sweetness profile, and compliance with phytosanitary and packaging requirements. Philippine mangoes maintain a premium positioning where exporters demonstrate reliability, quality discipline, and adherence to import standards.

Competitive Analysis

The global mango industry is highly competitive, with major exporters from Mexico, India, Thailand, Pakistan, and Brazil serving established international markets. Within this landscape, the Philippines differentiates through quality assurance standards, export compliance and discipline, and continued investment in supply chain efficiency.

Key competitive factors include:

Improved variety: The Company uses proprietary grafting of Carabao and Elena genetics to create a Dwarf Sweet Elena Carabao Mango variety suited for high-density planting. This new variety grows quickly, yields high-quality fruit, and is resilient. Sweet Elena is noted for being a large, exceptionally sweet variety of Philippine mango.

Quality differentiation: Consistent grading, structured post-harvest management, and origin branding initiatives enable Philippine exporters to access premium pricing tiers.

Compliance and certification: Success in high-value markets depends on meeting strict import regulations, including phytosanitary certifications and traceability requirements, and food safety documentation. Exporters with established compliance frameworks and quality management systems maintain preferential access to premium channels.

Supply chain integration: Efficient logistics, reliable cold chain infrastructure, and strategic partnerships with distributors and retailers are essential to maintaining product integrity and timely delivery. Post-harvest losses and quality degradation remain key competitive vulnerabilities.

Sustained competitiveness requires operational consistency, regulatory compliance, and the ability to respond to evolving market, sustainability, traceability and ethical sourcing.

Market Opportunities

Several growth drivers are emerging for the Philippine mango industry:

Value-added processing: Expansion into processed formats, including dried, puréed, or frozen forms, may diversify revenue streams and reduce exposure to fresh export volatility. Processed products typically command lower per-unit pricing but offer extended shelf life and reduced logistics complexity.

Emerging markets: Development of non-traditional export destinations in Europe, East Asia and Oceania offers potential for geographic diversification and demand expansion. These markets may require adapted packaging, sizing, or certification approaches.

Brand development: Strengthening origin labelling and sustainability credentials and varietal storytelling may support premium positioning and reinforce long-term market recognition. Consumer interest in provenance and ethical sourcing continues to grow across developed markets.

Sustainable practices: Investment in environmentally responsible cultivation practices, water management systems, and supply chain transparency will increasingly influence market access and premium pricing potential. Regulatory and consumer expectations regarding sustainability are rising across all major export markets.

Sustained growth will depend on disciplined market development, compliance alignment, infrastructure investment, and operational innovation within both fresh and processed export channels.

3. PRODUCTS AND SERVICES

Golden Forests provides investors with structured access to premium, high-value agroforestry assets positioned within global luxury and agricultural markets. Through professionally managed plantations in agarwood and premium mango varieties, the Company offers participation in two distinct but complementary asset profiles: long-cycle agarwood cultivation and income-generating mango production.

By integrating advanced scientific cultivation methods, proprietary inoculation techniques, and precision agricultural technologies, Golden Forests operates within a disciplined framework focused on quality, consistency, and long-term performance. Each investment is supported by defined contractual protections, transparent ownership structures, and professional plantation oversight aligned with international standards.

This approach positions agarwood and mango as strategically managed assets within a scalable and well-governed platform.

Client Services

Secure Ownership and Transparent Management

Golden Forests executes a sales and management contract with each client. This legally binding document establishes direct ownership of individually identified trees and defines the responsibilities for cultivation, maintenance, harvest, oud wood chip extraction, marketing and sales.

Official Owner Certificate

Upon completion of your purchase, each client receives an Official Owner Certificate, confirming the exact number of trees owned along with their precise GPS coordinates within the plantation. This provides verifiable traceability and asset identification within the managed plantation.

Comprehensive Tree Management and Replacement

All agarwood and mango trees are managed by experienced plantation teams to support optimal health and productivity. In the event of tree mortality or damage, replacement trees are allocated from a pre-established surplus planting reserve, matched to the corresponding maturity stage in accordance with the management agreement. The Company maintains a planting buffer designed to materially exceed projected mortality assumptions, supporting continuity of client holdings across the lifecycle of the asset.

Annual Plant Reports

Investors receive annual reports prepared by an independent and recognised auditing firm. These reports provide detailed updates on plantation performance, planting status and operational activity. Where replacement trees are allocated, corresponding GPS verification and asset adjustments are documented to maintain traceability and ownership transparency.

Investor Portal

Investors are provided with secure portal access to monitor asset records, plantation updates, and relevant documentation. The portal supports ongoing transparency and structured communication throughout the investment lifecycle, reinforcing accountability and clear reporting standards.

Flexible Asset Control

Investors retain the right to transfer, resell, or bequeath their tree holdings in accordance with the contractual and regulatory provisions governing the investment structure. This framework provides ownership flexibility while ensuring compliance with applicable legal and operational requirements.

Sustainable Impact Through Reforestation

For each agarwood and Carabao Mango tree acquired, Golden Forests allocates the planting of one native endemic species within designated reforestation areas. This one-to-one model links commercial plantation activity directly to regional ecosystem restoration and biodiversity support.

Client Plantation Visit Programme

As part of the investor engagement framework, Golden Forests provides access to a structured plantation site visit programme in Southern Zambales, Philippines.

The Company hosts a two-day operational visit, including guided plantation tours and management briefings, enabling clients to observe plantation operations and asset allocation processes firsthand.

All travel and accommodation arrangements, including those required to reach or depart from the hosted programme, remain the sole responsibility of the client.

Plantation Yield Expectations

This section presents a systematic methodology for yield optimisation and sustained plantation performance. The implementation of advanced cultivation practices and rigorous resource management ensures consistent output throughout each plantation asset's lifecycle.

Golden Forests' cultivation framework combines proven agronomic strategies, standard operating procedures, with proven agarwood inoculation and mango grafting techniques to ensure reliable yield results across both agarwood and mango plantations.

Detailed yield assumptions, lifecycle progression data, and productivity benchmarks are set out in Appendix D. All projections reflect proven cultivation practices and are subject to the assumptions and qualifications outlined in the Safe Harbour Statement.

Agarwood trees

As part of the plantation initiative, the Company harvests its clients' agarwood tree portfolio over a two-year period. This provides clients with a return on investment commencing in year nine from the date of out-planting and allows for a minimum period of two years from the date the trees were inoculated. Agarwood trees reach commercial maturity from year nine, with oud chip production continuing through year ten.

Each agarwood tree is projected to yield approximately 1.5 kgs of oud wood chips in its ninth year, increasing to approximately 2.0 kgs by year 10 (*see Appendix C*). These projections are derived from established cultivation and inoculation methodologies, developed in collaboration with the FIO and reflect conservative assumptions based on documented cultivation outcomes.

Actual yields are subject to variability based on tree health, inoculation timing, environmental conditions, and resin development patterns.

An illustration of the 10-year lifecycle and yield progression is presented in Appendix K. This appendix outlines each phase of growth and production, providing stakeholders with clear benchmarks for expected returns and operational planning.

Mango Trees

Mango trees are expected to commence fruit production on or before the fifth year, initially yielding approximately 30 kgs per tree. This output is anticipated to increase incrementally, stabilising at an annual yield of 100 kgs per tree from the tenth through the twenty-fifth year of the mango trees' productive lifespan (*see Appendix D*). Induced flowering techniques support year-round harvesting, reducing seasonal concentration risk and enabling consistent supply to market.

The Company uses proprietary grafting methods and dwarfing techniques that blend Carabao and Elena genetics to create a dwarf mango variety (*Dwarf Sweet Elena Carabao*), allowing high-density planting. This shift from traditional seasonal mango cultivation improves efficiency by enabling the use of agricultural drones for foliar fertilisation and pesticide spraying.

Yield assumptions reflect controlled growing conditions, structured irrigation management, and disciplined pest control protocols as described in Section 7 (*Operational Plan*).

Agarwood Products and Market Pricing

This section presents a detailed overview of the various commercial derivatives obtainable from a mature *Aquilaria crassna* agarwood tree. The analysis assumes a mature tree in its tenth year, weighing between 150 and 200 kgs, and outlines indicative price ranges per kg across principal derivative products.

For each derivative, the estimated price range, common commercial applications, and indicative yield considerations are presented to illustrate the potential value profile of a mature resin-producing tree.

Important disclaimer: The following indicative value ranges are provided for illustrative purposes only and are based on current market observations as of January 2025. Actual realised prices will vary materially according to resin quality, grade classification, product purity, geographic origin, processing methodology, market conditions at time of sale, and prevailing supply-demand dynamics. These ranges should not be relied upon as projections or guarantees of future returns. All pricing information is subject to the qualifications set out in the Safe Harbour Statement.

Product	Estimated Price Range (USD per kg)	Uses
Agarwood oil	10,000 - 50,000+	Perfumes, aromatherapy, traditional medicine, religious ceremonies
Agarwood essential oil	20,000 - 100,000+	High-end perfumery, aromatherapy, traditional medicine
Agarwood (oud) wood chips	750 - 5,000	Incense, crafting, aromatherapy, traditional medicine
Agarwood incense	500 - 2,000	Religious ceremonies, meditation, air freshening
Agarwood tea	600 - 3,000	Traditional medicinal purposes, relaxation, unique flavour
Agarwood hand cream	2,500 - 10,000	Skincare, moisturising, aromatherapy
Agarwood powder	600 - 3,000	Incense making, traditional medicine, cosmetics
Agarwood beads	200 - 1,000	Jewellery making, prayer beads, decorative items

Important Considerations

Price variability: All pricing ranges presented are indicative estimates based on prevailing international market observations and publicly available trade data. Actual realised prices may vary significantly depending on resin grade, product quality standards, authenticated origin, processing standards, and prevailing market demand at the time of sale. Market prices are subject to fluctuation based on global supply conditions, regulatory changes, and shifts in consumer preferences.

Market data and assumptions: The valuation ranges and commercial applications outlined above reflect widely recognised industry knowledge, publicly available trade information, and commonly accepted market trends within the agarwood sector. Pricing data is aggregated from trade publications, industry reports and market observations across multiple regions. For transaction-specific pricing or current market benchmarks, reference to specialised industry reports, trade associations, and subject-matter experts is recommended.

Yield variability: Estimated yield ranges are illustrative and may vary substantially according to tree health, genetic characteristics, environmental growing conditions, cultivation methodology, inoculation technique, and post-harvest processing methods. Actual production outcomes are subject to biological and operational variables inherent in plantation cultivation. Not all trees will produce equivalent resin volumes or quality grades.

Grade classification impact: Price realisation is heavily influenced by industry-standard grading systems. Super-grade and Grade A materials command significant premiums over lower grades. Grading assessments consider resin density, aromatic intensity, oil content, colour characteristics, and overall appearance. Golden Forests' production methodologies target consistent quality outcomes, but grade distribution across harvest volumes will vary.

Mango Products and Market Pricing

Mango trees provide a wide range of commercially valuable derivatives, with both the fruit and tree by-products utilised across food processing, cosmetics, and speciality ingredient markets. Below is a structured summary of common mango derivatives together with indicative value ranges.

Important disclaimer: The following indicative value ranges are provided for illustrative purposes only and are based on current market observations as of July 2025. Actual realised prices will vary materially according to product grade, quality standards, geographic origin, processing methodology, market conditions at time of sale, and prevailing supply-demand dynamics. These ranges should not be relied upon as projections or guarantees of future returns. All pricing information is subject to the qualifications set out in the Safe Harbour Statement.

Product	Description/Use	Estimated Value (USD)
Fresh mango fruit (domestic and export sales)	Major agricultural commodity, consumed domestically and exported; sold in local markets, grocery stores, roadside stands; exports to the US, EU, Middle East, East Asia	2.00 - 3.50 per kg (domestic); 4.00+ per kg (export, premium)
Mango pulp and juice	Used in beverages, purees, nectars, jams, baby foods	800 - 2,000 per tonne (bulk)
Dried mango (Amchur)	Produced by drying and powdering unripe mangoes; used as spice or souring agent	3,000 - 6,000 per tonne (premium retail grades)
Mango slices and chunks	Dried and canned; snacks or dessert ingredients	Dried: 4,000 - 8,000 per tonne; Canned: 1,200 - 2,500 per tonne
Mango pickles	Made from raw mangoes, pickled with spices and oils; popular in South Asian cuisine	1,500 - 3,500 per tonne
Mango seed kernel oil	Extracted from seed kernel; used in cosmetics, soaps, cocoa butter substitute	2,000 - 4,000 per tonne
Mango butter	Fat derived from mango seed; used in skincare products	3,500 - 5,500 per tonne
Mango peel powder	Drying and grinding peels; dietary fibre supplement, animal feed	500 - 1,200 per tonne (food grade)
Mango vinegar	Produced from fermented mango juice or pulp; condiment, preservative	1,000 - 3,000 per tonne (bottled retail product)
Mango wine	Fermented beverage, specialty markets	6 - 20 per bottle (750 ml)
Mango flour	Prepared from seeds or pulp; gluten-free baking, thickener	1,200 - 2,800 per tonne
Mango leaves and bark	Used in traditional medicine, tannins, fodder	300 - 800 per tonne (raw, dried material)

Mango waste Utilisation	Processing waste repurposed for animal feed, compost, extraction of antioxidants, pectin	100 - 600 per tonne (feed/compost); higher for extracted compounds

Market Context and Variability

The breadth of mango derivatives illustrates the economic versatility of the mango tree beyond fresh fruit sales. Value-added outputs support multiple commercial channels, including food processing, beverage production, cosmetics, and speciality consumer goods, such as pharmaceutical applications.

As with all agricultural commodities, realised pricing, quality, and yield may vary substantially depending on mango cultivar, fruit maturity at harvest, processing methods, supply conditions, seasonal factors, and prevailing market demand across different product categories.

Fresh fruit sales represent the primary revenue stream for most commercial mango operations, with value-added processing serving as supplementary or opportunistic revenue where processing infrastructure, market relationships, and economies of scale justify the investment.

Note: All values presented are indicative estimates based on general market trends and publicly available industry information as of July 2025. For current transaction-specific pricing, consultation with specialised market reports, industry trade associations, and subject-matter experts is recommended. Golden Forests' initial commercial focus is on premium fresh fruit exports, with value-added processing opportunities under evaluation for future consideration.

4. BUSINESS MODEL AND REVENUE STREAMS

Business Model Overview

Golden Forests operates a capital-efficient agroforestry platform that generates revenue through the sale and lifecycle management of high-value biological assets. The Company cultivates premium agarwood and Carabao Mango trees, selling individual trees to private and professional clients whilst retaining long-term management responsibilities and commission-based revenue from harvest sales.

The model creates a self-funding growth mechanism: upfront tree sales finance plantation operations and expansion, whilst deferred commission revenue provides recurring cash flow over the 10–25-year asset lifecycle. The Company’s role as exclusive plantation manager, processor, and export facilitator ensures control over quality, client satisfaction, and margin capture throughout the value chain.

Scalability and Geographic Expansion Profile

Replicable Business Model Across Geographies and Crops

Golden Forests has intentionally designed its business model to be easily replicated in different regions and with various crop types. The current operations in the Philippines serve as a proof of concept, demonstrating that the

operational framework is effective and can be adapted to other locations. This means that the systems, processes, and strategies developed are not unique to one place or crop; they are built to be flexible and scalable, allowing the company to expand into new markets with minimal adjustments.

Significance:

- **Risk diversification:** By not being tied to a single geography or crop, the company can reduce risks associated with local disruptions (e.g., weather, regulations, pests).
- **Growth potential:** The ability to replicate the model means faster and more efficient expansion into new markets.

Adaptable Technology and Methodologies

The Agroforestry Intelligence platform, along with advanced cultivation methods and a strong client management system, is designed for adaptability. These systems can be applied to:

- Other Southeast Asian and South Asian regions that possess suitable agro-climatic conditions for successful agarwood cultivation.
- Geographies where there is an established mango export infrastructure or where emerging markets for high-value fruit are present.
- Additional premium agroforestry crops that exhibit similar long-cycle value appreciation and show strong demand in export markets.

Significance:

- **Technology transfer:** The platform and methods are not locked to one crop or region, making it easier to train new teams and onboard new partners.
- **Market responsiveness:** The company can quickly pivot to crops or regions where there is growing demand or better profitability.

Systematic and Efficient Expansion

Annual plantation expansion is carefully managed to ensure that each new hectare adds only minimal incremental overhead. This efficiency is achieved through:

- Centralised management systems that streamline decision-making and resource allocation.
- Implementation of standard operating procedures and engagement of scientific experts.
- Established supply chains that reduce logistical complexity and costs.
- Technology-enabled monitoring infrastructure that allows for remote oversight and rapid response to issues.

Significance:

- **Cost Control:** By keeping overhead low, the company can scale up without proportionally increasing expenses.

- **Operational Coherence:** Centralised systems ensure that standards and best practices are maintained as the company grows.
- **Scalable Growth:** The infrastructure supports expansion without sacrificing quality or efficiency.

Focus on High-Value, Long-Cycle Crops

The company concentrates on high-value agroforestry crops that gain value over extended periods and are sought after in international markets. By doing so, it guarantees investments in crops likely to deliver substantial profits and consistent demand.

Significance:

- **Sustainable profitability:** Long-cycle crops can provide steady income streams and capitalise on market trends.
- **Export opportunities:** Growing crops for export expands markets and lowers dependence on local buyers.

In summary

Golden Forests' approach is built for growth, flexibility, and efficiency. By leveraging adaptable technology, systematic expansion processes, and a focus on high-value crops, the company is well-positioned to scale its operations across new regions and markets while maintaining cost-effectiveness and operational excellence.

Revenue Model Structure

The Company generates revenue through two primary mechanisms:

1) Upfront Tree Sales (capital inflow)

   Clients purchase individual trees at fixed per-tree prices that incorporate all lifecycle costs: land lease and preparation, infrastructure development, irrigation equipment, nursery setup, seedlings, planting, ongoing tree care, monitoring and security, maintenance, inoculation (agarwood), harvesting, extraction, processing, export logistics, and sales.

   Tree sales provide immediate capital to fund all operational requirements and continued plantation expansion. As plantation acreage scales annually, projected at 10% growth per annum (*see Appendix A*), upfront sales revenue creates a self-reinforcing growth cycle.

   The current pricing structure and volume discount framework are outlined in Section 6 (*Marketing and Sales Strategy*). Detailed financial assumptions, including tree price escalation factors, revenue growth projections, and comprehensive financial modelling, are provided in Section 9 (*Financial Plan and Projections*).

2) Commission Revenue (recurring cash flow)

Upon harvest, the Company processes, exports, and sells biological output on behalf of tree owners, retaining commission on gross sales:

- **Agarwood oud wood chips:** 10% commission on export sales (years 9-10)
- **Mango fruit:** 20% commission on domestic and export sales (years 5-25)

This deferred commission structure aligns Company incentives with client outcomes whilst generating high-margin recurring revenue as plantation cohorts mature sequentially. Commission revenue requires minimal incremental cost once plantations are established, supporting strong EBITDA margins as operations scale.

3) Additional Revenue:

- Interest income from operation bond holdings
- Potential value-added processing revenue from mango by-products (under evaluation)
- Potential government grants, agriculture development incentives, or sustainability programme participation (subject to regulatory approval and eligibility)
- Future carbon credit or ecosystem services revenue linked to the reforestation programme (under evaluation)

Client Return Profile - Evidence of Business Model Viability

The strength of Golden Forests' business model is demonstrated by the compelling return profile offered to tree purchasers. Strong client returns drive demand, enable premium pricing, and support the Company's ability to scale tree sales whilst maintaining healthy commission margins.

Agarwood Client Returns

Agarwood tree owners receive proceeds from oud wood chip sales (net of 10% commission) commencing in year nine (50% of holdings harvested) and year ten (remaining 50% harvested). Projected annualised return is approximately 18% over the 10-year holding period (*see Appendix B).*

Mango Client Returns

Mango tree owners receive proceeds from fruit sales (net of 20% commission) commencing in year 5, with annual harvests continuing through year 25. Projected annualised return is approximately 12% for years 1–15, increasing to over 20% for years 16–25 as trees reach peak production maturity (*see Appendix B*).

Strategic Significance

These return profiles create multiple business advantages:

- **Organic demand generation**: Strong client returns lead to referrals and repeat business with new plantation groups.
- **Premium pricing power:** Strong performance enables the Company to maintain pricing discipline whilst delivering competitive client outcomes.

- **Market validation**: Demonstrated client returns provide credibility for geographic expansion and new crop introduction
- **Scalability Support:** satisfied clients become advocates, reducing customer acquisition costs and supporting efficient growth

Detailed client return assumptions, illustrative investment examples, harvest yield projections, and comprehensive financial modelling are provided in Sections 9 (*Financial Plan and Projections*) and 10 (*Investment Highlights*).

Revenue Risk Mitigation

The Company maintains structured programmes to protect both client satisfaction and the Company's revenue stability:

Mortality protection: Surplus trees (20% buffer stock) are planted annually to provide replacement inventory for mortality events caused by lightning strikes, cyclones, typhoons, drought, floods, pests, diseases, animal damage, theft, or vandalism. Replacement trees are allocated from surplus inventory and matched to the maturity stage of affected trees. This framework protects client holdings and maintains Company commission revenue projections by ensuring harvest timelines remain intact.

Quality assurance: All plantations are managed in accordance with structured cultivation protocols, supported by Agroforestry Intelligence monitoring systems and scientific partnerships with leading Philippine agricultural research institutions and scientific experts, as provided in Section 8 (*Company Organisation*).

Regulatory compliance: The Company maintains CITES accreditation for agarwood import/export and is pursuing forest certification (FSC for agarwood, PEFC for mango) to strengthen market access and support premium positioning in international markets (*see Appendix H*).

Comprehensive risk assessment across environmental, socio-economic, and natural hazard categories, together with detailed mitigation strategies, are outlined in Section 11 (*Risk Assessment and Mitigation Strategies*).

Strategic Value for Company Investors

This business model creates multiple competitive advantages:

Self-funding growth: Tree sales finance expansion without requiring continuous external capital raises, enabling disciplined scaling whilst preserving equity dilution and maintaining financial flexibility.

Aligned incentives: Company profitability depends directly on client satisfaction and harvest success, ensuring operational excellence and quality focus throughout the cultivation and export value chain.

Diversified revenue timing: Cohorts planted across multiple years create staggered commission revenue streams, smoothing cash flow volatility and reducing dependency on single harvest cycles or seasonal concentration.

High-margin recurring revenue: Commission revenue from mature plantations generates predictable cash flow with minimal incremental costs, supporting strong profitability metrics as operations scale across multiple plantation cohorts.

Long-term client relationships: The 10–25-year asset lifecycle creates deep client engagement, organic referrals through demonstrated performance, and potential for repeat purchases across new plantation cohorts or geographic expansions.

Geographic and crop optionality: The proven business model architecture can be replicated across suitable regions and extended to additional high-value agroforestry crops, creating multiple expansion pathways without fundamental restructuring of operational or commercial frameworks.

Capital efficiency: The Company's ability to generate upfront revenue through tree sales, whilst deferring major operational costs through bond-funded prepaid expense structures, creates favourable working capital dynamics and reduces reliance on debt financing during plantation establishment phases.

The Company's integrated approach, combining biological asset cultivation, professional lifecycle management services, export market access, and structured client participation, positions Golden Forests as a scalable platform capable of capturing value across the premium agroforestry sector whilst maintaining disciplined capital efficiency and operational control.

5. MARKETING AND SALES STRATEGY

Golden Forests employs a multi-channel marketing and sales strategy to attract private and professional clients seeking sustainable, asset-backed agroforestry participation. The Company's approach combines digital marketing infrastructure, professional agent partnerships, and structured client engagement to build a diversified client base across geographic markets.

Target Clients

The Company's target clients for tree purchase opportunities are private and professional clients seeking sustainable, impact-driven investment opportunities. These tree purchases are positioned either as alternative assets or as part of broader portfolio diversification strategies, offering tangible, asset-backed exposure to emerging market agriculture.

Target segments include:

Private Clients

- High-net-worth individuals seeking alternative asset exposure
- Family offices incorporating ESG or impact-oriented allocations
- Environmentally conscious individuals combining financial objectives with measurable sustainability outcomes

Professional Clients

- Institutional investors with sustainable agriculture or impact mandates
- Corporate entities seeking carbon offset or sustainability programme participation
- Fund managers building alternative asset portfolios with tangible backing

The Company also selectively engages retail clients who meet the required financial profile and demonstrate the long-term investment horizon appropriate for agroforestry assets, and who value measurable environmental impact.

Client Value Proposition

Golden Forests offers clients the opportunity to acquire individually identified trees within professionally managed plantations, combining:

- **Tangible asset ownership:** Direct ownership of GPS-verified biological assets with transparent traceability
- **Lifecycle management:** Comprehensive cultivation, harvesting, processing, and export services are included in the upfront purchase price
- **Projected returns:** Target annualised returns of 18% (agarwood, 10-year cycle) and 12-20% (mango, 25-year cycle) (*see Appendix B*)
- **Sustainability impact:** One-to-one native species reforestation linked to each tree purchased, supporting ecosystem restoration
- **Professional oversight:** Operations managed through Agroforestry Intelligence platform with scientific partnership support
- **Mortality Protection:** 100% tree replacement guarantee for specified adverse events
- **Operational Transparency:** Annual reporting, plantation visit programme, and secure client portal access

This value proposition differentiates Golden Forests within the sustainable agriculture and alternative asset sectors by combining institutional-grade operational standards with accessible participation structures and measurable impact outcomes.

Distribution and Sales Channels

Professional Agent Network

Golden Forests engages professional sales agents and third-party consultants across key geographic markets to support client acquisition and relationship management. Agent partners are selected based on:

- Established networks within target client segments
- Regulatory compliance and professional standing
- Experience in alternative assets, sustainable investments, or agricultural products
- Geographic market knowledge and cultural alignment

Agents operate under structured agreements with defined commission structures, compliance requirements, and performance expectations. The Company provides comprehensive sales support materials, product training, and ongoing technical assistance to ensure consistent client messaging and service quality.

Current agent network coverage includes:

- Middle East (United Arab Emirates, Saudi Arabia)
- Europe (United Kingdom, Germany)
- Southeast Asia (Malaysia, Singapore)
- Expanding presence in additional target markets, subject to regulatory approval and partnership identification

Digital Marketing Infrastructure

The Company operates an integrated digital marketing platform combining:

- **Website and client portal:** Centralised information hub, client account management, and secure document access
- **Content marketing:** Educational materials on agroforestry, sustainability impact, and market dynamics
- **Digital advertising:** Targeted campaigns across search, social media, and specialist investment platforms
- **Email marketing:** Segmented communications based on client interests, engagement levels, and purchase stage
- **CRM and lead management:** Structured pipeline management, tracking prospects from initial inquiry through client conversion

To maximise reach and effectiveness, we actively engage professional agents and third-party consultants to outsource marketing and sales efforts. These agents leverage their established networks and market expertise to identify and connect with prospective clients, ensuring broad exposure across local and global markets. By partnering with skilled sales agents, we enhance our marketing initiatives, streamline the client acquisition process, and offer comprehensive support to clients throughout their decision-making journey.

AI-Enabled Marketing Optimisation

Golden Forests employs data analytics and machine learning technologies to enhance marketing efficiency and client engagement:

- **Audience segmentation:** Behavioural analysis and preference mapping to identify high-potential prospect segments
- **Personalised messaging:** Automated delivery of tailored content based on prospect profile, engagement history, and expressed interests
- **Campaign performance analysis:** Real-time monitoring and optimisation of digital advertising spend and channel effectiveness
- **Lead scoring:** Predictive models identifying marketing-qualified leads (MQLs) ready for sales team engagement

Product Marketing Capabilities

In addition to client acquisition marketing, Golden Forests is developing marketing infrastructure to support the sale of harvested products (agarwood oud wood chips and mango fruit) on behalf of clients. This includes:

- Market intelligence and pricing analysis for premium agarwood and mango export markets
- B2B relationship development with international buyers, processors, and distributors
- Product positioning, quality certification documentation, and origin branding
- Digital marketplace presence and e-commerce capabilities for direct sales channels

This dual marketing capability, client acquisition and product commercialisation, creates operational synergies and supports the Company's commission revenue model by maximising realised prices for client harvest outputs.

Pricing Structure

Standard Pricing (introductory rates)

- Agarwood**:** USD 292.50 per tree
- Carabao Mango: USD 592.50 per tree

Per-tree pricing includes comprehensive lifecycle services

- Land lease and preparation
- Infrastructure development and nursery setup
- DNA-verified seedlings (agarwood) or grafted cultivars (mango)
- Planting and establishment
- Ongoing tree care, monitoring, and maintenance
- Inoculation services (agarwood, years 7-8)
- Harvesting, processing, and quality grading
- Export logistics and international sales

Volume Discount Structure

A tiered discount framework is offered for volume purchases:

AGARWOOD	Price Per Tree (USD)	Min Trees
Institutional Investors		
6 ha = 14.86 ac = 10,000 trees	242.50	10,000
3 ha = 7.41 ac = 5,000 trees	247.50	5,000
2 ha = 4.94 ac = 3,334 trees	252.50	3,334
1 ha = 2.47 ac = 1,667 trees	262.50	1,667
Private Investors		
1,000-2,000 trees	262.50	1,667
1,000 trees	262.50	1,000
500-1,000 trees	272.50	500
200-500 trees	282.50	200

100-200 trees	292.50	100
50 trees	292.50	50
25 trees	292.50	25

MANGO	Price Per Tree (USD)	Min Trees
Institutional Investors		
6 ha = 14.86 ac = 2496 trees	542.50	2496
4 ha = 9.88 ac = 1664 trees	547.50	1664
3 ha = 7.41 ac = 1248 trees	547.50	1248
2 ha = 4.94 ac = 832 trees	552.50	832
1 ha = 2.47 ac = 416 trees	562.50	416
Private Investors		
200-500 trees	572.50	200
150 trees	592.50	150
100 trees	582.50	100
50 trees	592.50	50
25 trees	592.50	25

Annual Price Escalation

Tree prices are subject to annual adjustment to reflect input cost inflation and operational expenses:

- Agarwood: 1.5% annual escalation
- Mango: 3.0% annual escalation

Harvest Revenue Assumptions

Projected client proceeds are based on conservative market pricing assumptions with 1.5% annual escalation:

- Mango fruit: USD 3.00/kg (domestic), USD 4.00/kg (export)
- Agarwood oud wood chips: USD 775/kg (mid-premium grade)

For an in-depth analysis of current market prices and future pricing trends, see Appendix E.

Commission Structure

The Company retains commission on harvest sales:

- Agarwood: 10% of gross oud wood chip sales
- Mango: 20% of gross fruit sales

Clients receive net proceeds after commission deduction. All pricing assumptions and projected returns are detailed in Section 9 (*Financial Plan and Projections*) and are subject to the qualifications outlined in the Safe Harbour Statement.

Marketing and Sales Performance Metrics

The Company tracks key performance indicators across the client acquisition funnel:

Marketing Metrics

- Lead generation volume and cost per lead
- MQL conversion rates by channel and campaign
- Content engagement and digital advertising ROMI
- Geographic and demographic reach

Sales Metrics:

- MQL to SQL conversion rates
- SQL to client conversion rates
- Average transaction value and trees per client
- Sales cycle duration
- Client acquisition cost

Agent Performance:

- Sales volume by agent and geography
- Commission payout vs revenue generated
- Client satisfaction and retention rates

Customer lifetime value (CLV)

Metric	Value (15 Years)
Total Net Sales/Revenue	USD 574.9M
Total Net Income After Tax	USD 321.8M
Total Trees (Agarwood and Mango)	1,588,623
Average Revenue per Tree	USD 362
Average Net Income per Tree	USD 203
Estimated CLV (per tree)	**USD 203**

Client acquisition cost (CAC)

Item	Amount (over 15 years)
Marketing	USD 36.9 M
Sales Commission	USD 147.8M
Land Leasing Incentive	USD 12.9M
Tree Discount	USD 73.9M
Total Acquisition Cost (15 years)	USD 271.6 M
Acquisition Cost per Tree	**USD 171**

Return on marketing investment (ROMI) by channel (Example: 5 years)

Year	Gross Profit (USD)	Marketing & Sales Budget (USD)	Sales Commission (USD)	Total Marketing Investment (USD)	ROMI
2026	11.8M	898.3M	3.82M	4.72M	1.50
2027	13.1M	1.0M	4.30M	5.37M	1.44
2028	14.6M	1.2M	4.83M	6.04M	1.41
2029	16.2M	1.35M	5.43M	6.79M	1.39
2030	18.1M	1.52M	6.11M	7.63M	1.37

ROMI values decrease slightly over time as marketing investments increase, but remain above 1.3, meaning every dollar spent on marketing returns USD 1.3 – USD 1.5 in gross profit.

These metrics inform ongoing optimisation of marketing spend allocation, agent partnership management, and sales process refinement to support efficient scaling of client acquisition activities.

Client plantation visit programme

Golden Forests provides clients with the opportunity to participate in a structured plantation site visit programme in Southern Zambales, Philippines. The programme is designed to support transparency, asset verification, and informed client engagement.

The Company offers a hosted two-day operational site visit, which includes guided plantation inspections, technical briefings, and meetings with management and agronomic teams, and hosted accommodation and meals during the scheduled visit period.

Travel to and from the Philippines, as well as any accommodation, travel, or personal expenses outside the hosted two-day programme, remain the responsibility of the client.

Additional Visits

Clients may arrange additional plantation visits, subject to prior coordination with the Company. Costs associated with additional visits beyond the hosted programme are the responsibility of the client unless otherwise agreed.

The plantation visit programme is designed to strengthen client confidence, provide operational transparency, and reinforce the tangible nature of tree ownership within professionally managed estates.

7. OPERATION PLAN

Location and Strategic Rationale: Zambales, Philippines

Global Forests' initial plantation operations are established in Zambales Province, selected for its favourable agro-climatic conditions, established agricultural infrastructure, and strategic accessibility.

Agro-Climatic Advantages

Zambales features a tropical climate with distinct wet and dry seasons, optimal for both *Aquilaria crassna* agarwood and Carabao Mango cultivation:

- **Rainfall distribution:** Annual precipitation of 1,500–2,500mm with a pronounced dry season (October–June) supporting flower induction in mango and an appropriate moisture balance for agarwood development
- **Temperature range:** Consistent 24–32°C temperatures throughout the growing season support steady biological development without thermal stress
- **Seasonal patterns:** Extended dry period benefits mango fruit development and harvesting operations, whilst wet season replenishment supports water table maintenance and irrigation planning
- **Soil Characteristics:** Well-drained volcanic soils with appropriate pH ranges (5.5–7.5) and nutrient profiles for both species

Agricultural Heritage

Zambales is recognised for producing some of the Philippines' highest-quality Carabao Mangoes, demonstrating proven suitability for premium mango cultivation. This established agricultural sector provides**:**

- Access to experienced plantation labour with mango cultivation expertise
- Established supply chains for agricultural inputs and services
- Proven cultivation protocols and best practice frameworks
- Local knowledge of pest management and seasonal optimisation

Infrastructure and Accessibility

The province provides operational advantages supporting efficient plantation management:

- **Proximity to Clark International Airport:** Approximately 1.5 hours' drive, facilitating international client visits, management travel, and potential air freight for high-value products
- **Distance from Manila:** Four hours by road, providing access to capital city services, export logistics, and financial/legal infrastructure
- **Established agriculture sector:** Zambales is recognised for premium Carabao Mango production, providing access to experienced agricultural labour, established supply chains, and proven cultivation expertise
- **Port access:** Coastal location supports efficient export logistics for containerised shipments to international markets

Geographic Scalability

Whilst Zambales serves as the Company's initial operational base, the cultivation model is designed for geographic replication. Site selection criteria and operational frameworks developed in Zambales can be applied to suitable locations across Southeast Asia and South Asia, supporting the Company's long-term expansion strategy.

Climate Overview: Zambales

Historical Climate Overview

Zambales experiences a tropical climate characterised by distinct seasonal patterns, optimal for agroforestry operations:

- **Wet season (July–September):** Peak rainfall supporting plantation irrigation requirements and water resource replenishment
- **Dry season (October–June):** Extended period with minimal precipitation, critical for mango flowering induction and fruit development
- **Temperature stability:** Year-round warm temperatures with minimal seasonal variation, supporting continuous growth cycles
- **Typhoon exposure:** Southern Zambales experiences lower typhoon frequency compared to northern Philippine regions, reducing catastrophic weather risk

Historical Climatic Data: Zambales Province, Philippines

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Nov	Oct	Dec	Year
Record high °C (°F)	33.0 (91.4)	35.0 (95.0)	36.0 (96.8)	38.0 (100.4)	36.0 (96.8)	37.0 (98.6)	33.0 (91.4)	33.0 (91.4)	32.0 (89.6)	33.0 (91.4)	33.0 (91.4)	33.0 (91.4)	38.0 (100.4)
Average high °C (°F)	29.04 (84.27)	29.93 (85.87)	31.13 (88.03)	32.23 (90.01)	32.04 (89.67)	30.25 (86.45)	28.99 (84.18)	28.7 (83.66)	29.35 (84.83)	29.66 (85.39)	30.23 (86.41)	29.66 (85.39)	30.1 (86.18)
Daily mean °C (°F)	26.23 (79.21)	26.83 (80.29)	28.03 (82.45)	29.22 (84.6)	29.66 (85.39)	28.38 (83.08)	27.34 (81.21)	27.17 (80.91)	27.46 (81.43)	27.43 (81.37)	27.7 (81.86)	27.09 (80.76)	27.71 (81.88)
Average low °C (°F)	21.94 (71.49)	22.05 (71.69)	23.06 (73.51)	24.48 (76.06)	25.59 (78.06)	25.23 (77.41)	24.45 (76.01)	24.63 (76.33)	24.4 (75.92)	23.86 (74.95)	23.74 (74.73)	23.3 (73.94)	23.89 (75.0)
Record low °C (°F)	18.0 (64.4)	19.0 (66.2)	20.0 (68.0)	21.0 (69.8)	23.0 (73.4)	22.0 (71.6)	21.0 (69.8)	18.0 (64.4)	18.0 (64.4)	19.0 (66.2)	18.0 (64.4)	20.0 (68.0)	18.0 (64.4)
Average precipitation mm (inches)	14.9 (0.59)	14.4 (0.57)	27.33 (1.08)	56.91 (2.24)	131.2 (5.17)	319.14 (12.56)	346.75 (13.65)	377.53 (14.86)	250.25 (9.85)	186.78 (7.35)	64.75 (2.55)	46.38 (1.83)	153.03 (6.02)
Average precipitation days (≥ 1.0 mm)	4.45	3.73	7.09	13.18	19.64	27.27	27.91	25.55	24.64	19.91	9.18	8.45	15.92
Average relative humidity (%)	74.09	71.51	71.66	71.96	74.54	81.02	84.2	84.21	83.56	81.66	77.04	76.47	77.66
Mean monthly sunshine hours	8.39	10.58	11.16	12.5	12.51	11.87	11.33	11.1	10.8	10.48	9.13	8.26	10.68

Plantation Site Selection Criteria

Golden Forests applies structured site assessment protocols, ensuring optimal growing conditions:

Criterion	Agarwood	Mango
Climate	Tropical with high humidity and consistent moisture. Annual rainfall >1,500mm, average temperatures 20–25°C. Dry season of up to four months tolerable.	Tropical, seasonally wet/dry climate with distinct three-to-five-month dry period to trigger flowering and fruiting.
Soil	Well-drained, fertile loam or sandy-clay loam soil with a slightly acidic pH of 5.5–7.5.	Adaptable to sandy, loam, and clay, provided they are deep and well-drained. pH of 5.5–7.5 preferred.
Topography	Moist, well-drained slopes or areas along streams, avoiding waterlogged conditions.	Flat to rolling lands with good drainage. Deep soil to accommodate its extensive root system.
Sunlight	Full sun exposure. Young plants can tolerate some shade, but older trees need more light.	Abundant sunshine, especially for flowering and fruit development. Partial shading severely reduces productivity.

Technology Infrastructure and Plantation Monitoring Systems

Golden Forests deploys an integrated suite of precision agriculture technologies across all plantation operations. These systems form the operational backbone of the Agroforestry Intelligence framework, providing real-time monitoring, predictive analytics, and decision support throughout the cultivation lifecycle.

AI-Enabled Monitoring Systems

Across Golden Forests' plantations, AI-enabled systems support:

- **Soil condition monitoring:** Real-time tracking of moisture levels, nutrient balance, pH analysis, and soil structure integrity through ground-based sensor networks
- **Climate and micro-weather tracking:** Localised weather pattern analysis to inform irrigation scheduling, stress management, and harvest window identification
- **Early detection systems:** Pattern recognition algorithms identify pest infestations, disease outbreaks, and plant stress indicators before visible symptoms emerge, enabling proactive intervention
- **Growth tracking and yield forecasting:** Predictive analytics models monitor tree development, estimate maturity timelines, and forecast production volumes based on biological indicators
- **Harvest optimisation:** Data-driven determination of optimal harvest timing based on maturity indicators, weather windows, and market conditions

This data infrastructure enables precision resource allocation, reducing water consumption, minimising chemical inputs, and lowering labour costs whilst improving overall plantation health and productivity.

Drone-Supported Aerial Analysis

Agricultural drones are deployed for comprehensive aerial surveillance and mapping across plantation estates. Drone technology enables:

- **Large-scale crop health assessment:** Multispectral imaging identifies stress patterns, nutrient deficiencies, and irrigation inconsistencies across extensive areas
- **Early stress identification:** Thermal imaging detects temperature variations indicating water stress, disease activity, or root zone problems before ground-level observation would reveal issues
- **Targeted input application:** Precise mapping guides fertiliser, pesticide, and soil amendment application, reducing waste and environmental impact whilst improving cost efficiency
- **Yield estimation modelling:** Pre-harvest aerial analysis supports production forecasting and logistics planning, enabling better market timing and resource allocation

Data collected through drone operations is integrated directly into the Agroforestry Intelligence platform, with analysis outputs feeding into irrigation schedules, maintenance protocols, and harvest planning systems.

Smart Irrigation and Soil Sensor Networks

Ground-based sensor networks provide continuous monitoring of soil conditions at multiple depths across plantation zones. Smart irrigation systems automatically adjust water delivery based on:

- Current soil moisture levels and plant water uptake rates
- Weather forecasts and evapotranspiration modelling
- Growth stage requirements and seasonal patterns
- Water table depth and groundwater availability

Integrated Pest Management Systems

AI-powered pest monitoring combines visual recognition systems, pheromone trap data, and environmental condition tracking to predict and prevent infestations. The system enables:

- Early intervention before pest populations reach economically damaging levels
- Targeted application of biological controls or minimal chemical inputs, reducing environmental impact
- Documentation of pest pressure patterns to inform future planting decisions and varietal selection
- Compliance with international phytosanitary requirements for export markets, supporting premium market access

Inoculation Technology and Resin Development Tracking (Agarwood)

For agarwood cultivation specifically, the Company employs proven proprietary inoculation formulas and techniques designed by the FIO. Post-inoculation monitoring systems track:

- Infection site development and resin formation progression
- Tree health indicators and stress responses to inoculation
- Optimal timing for secondary inoculation interventions
- Resin quality indicators to guide harvest window decisions and grading expectations

This structured approach to inoculation management supports consistent resin quality and maximises recoverable oud wood chip volumes per tree, directly impacting client returns and Company commission revenue.

Satellite Geotagging and Traceability Systems

Every tree within Golden Forests' plantations is assigned GPS coordinates, creating a comprehensive digital map of plantation assets. This geotagging infrastructure supports:

- Individual tree ownership verification for client portfolios
- Mortality tracking and replacement tree allocation in accordance with guaranteed provisions
- Harvest documentation and chain-of-custody requirements for export compliance
- Compliance with international timber tracking and anti-illegal logging frameworks (particularly relevant for CITES-regulated agarwood)

The integration of agronomic expertise, AI-enabled monitoring, drone-supported analysis, and precision irrigation systems positions Golden Forests to deliver consistent yields, operational efficiency, and transparent reporting throughout the plantation lifecycle.

Operational Workflow and Management Structure

Nursery Operations

- Species-verified seedling propagation (*Aquilaria crassna* DNA authentication)
- Grafted cultivar development (Dwarf Sweet Elena Carabao Mango hybrid)
- Quality control and health screening before out-planting
- Staged production to align with annual plantation expansion schedules

Plantation Establishment

- Land preparation, soil amendment, and irrigation infrastructure installation
- GPS-verified tree planting with individual identification and client allocation
- Intercropping protocols (cassava for agarwood shade provision in years 1-2)
- Buffer stock planting (20% surplus) for the mortality replacement programme

Ongoing Cultivation Management

- Scheduled fertilisation, pruning, pest monitoring, and disease prevention protocols
- Irrigation and fertigation management through smart sensor networks
- Pruning, canopy management, and spacing optimisation
- Quality inspections and AI-monitored health assessments

Harvest and Post-Harvest Operations

Agarwood (Years 7-10):

- Inoculation (years 7-8) using proprietary formulations
- Harvest (years 9-10) based on resin development monitoring
- Oud wood chip extraction (years 9-10), grading, and quality certification
- Export preparation and international logistics coordination

Mango (Years 5-25):

- Year-round flowering management for continuous harvest capability
- Quality grading and post-harvest handling protocols
- Cold chain logistics for domestic and export markets
- Value-added processing options (under development)

All operational activities are conducted in accordance with planned forest certification standards (FSC for agarwood, PEFC for mango) and international export requirements (*see Appendix H*)

8. COMPANY ORGANISATION

Corporate Organisation

Golden Forests operates through a multi-jurisdictional structure designed to optimise operational efficiency, regulatory compliance, and market access:

Golden Forests ADGM Holding Company (Abu Dhabi, UAE)

- Group-level governance and strategic oversight
- Capital structuring and client relations
- Financial management and fund administration
- Registered Office: Level 25, Al Sila Tower, Al Maryah Island, ADGM, Abu Dhabi, UAE

Golden Forests Agroforestry Intelligence Investment in Agricultural Enterprises & Management FZCO (Dubai, UAE)

- International sales and marketing operations
- Client relationship management and business development
- Regional market coordination (Middle East, Europe, Southeast Asia)
- Registered Office: IDZA Business Park, DDP 75773-001, A1-3641379065, Dubai Digital Park, Dubai Silicon Oasis, Dubai, UAE

Crassna Agroforestry Development Incorporated (CADI) (Philippines)

- Plantation operations and asset management
- Cultivation, harvesting, processing, and export logistics
- Local regulatory compliance and community engagement
- Registered Head Office: Level 24, Philippines Stock Exchange Tower, One Bonafacio Street, 5th Ave Cor.28th Street, BGC, Taguig City, Philippines

This structure provides clear separation between Group governance, international client engagement, and operational execution, whilst maintaining integrated strategic direction and quality standards across all entities.

Executive Management

CHAIRMAN

CHARLES MCKENZIE

An experienced executive of over 30 years, mostly based in London, specialising in Asset Management-Fixed Income investments.

Charles began his career with Deutsche Morgan Grenfell. During a 12-year engagement, he spent 3 years in Frankfurt and 1 year in Sydney as portfolio manager and Head of Fixed Income.

-Spent 7 years with Aberdeen Asset Management, Head of Fixed Income EMEA before taking on the role of Managing Director for JP Morgan Asset Management for 2 years. In 2014, Charles was appointed Chief Investment officer, Fixed Income for Fidelity International.

Charles holds a BSc Hons in Economics from Southampton University, and an MBA (Finance) from London City University Business School.

CHIEF EXECUTIVE OFFICER

MARK QUINN

-Over 25 years experience with alternative fuels , Agro Forestry, Agri Business & renewable energy generation in the developing markets

Spent 5 years in India, conducting research and development of various tree and grass species and application of biotechnology and microbiology .

Implemented a 300-hectare biomass tree project for a major Philippine Corporate for supply of biomass for power generation.

Established Essential oil distillation facilities in the Philippines for Ylang Ylang, Vetiver and Elemi Oils. Responsive for large scale cultivation of Vetiver grass.

Introduced and supplied the first Biofertilizer to the Philippines Government produced from nanotechnology.

Secured first CITIES import permits for Agarwood. 5 years of experience of implementing agarwood plantations in the Philippines

MARKETING AND SALES OFFICER

CORD KABUS-DUPRÉE

+20+ years financial advisory and family office experience and holds a Masters Degree in Business Administration and CFP Certified Financial Planner.

Established relationships across European and Middle Eastern wealth management centers with expertise in alternative asset investor marketing and institutional capital development.

Cord is highly recognized and successful senior sales and key account/client management professional . His experience combines a consulting sales approach with key business wide skills to transform opportunities into profitable revenue streams.
He spent 12 years with Horbach Wirtschaftsberatung GmbH (Swiss Life Group Germany), where he gained continued recognition for his financial, investment and IT skills, leading to his appointment as a Partner .

Over the past 5 years, Cord has been engaged in the marketing and sales of healthcare products in the US, and responsible for the supply of key medical equipment supplies during the Covid-19 epidemic.

FINANCIAL CONTROLLER

ANGIE BRION

-Experienced Accountant and Finance Controller with more than 20 years of quantifiable achievements and holds a Batchelor of Science Degree in Accounting

-An expertise in tax preparations, audit and financial modelling, with strong qualifications in all area of accounting, that produce bottom-line results and financial strength.

Angie has experience with the Philippine Board of Investments (BOI) and has successfully secured tax incentives and waivers for two SEC registered companies.

-Strong background and knowledge of Philippine Bureau of Customs Importation requirements and Securities and Exchange compliance

A specialist in Environment of Department of and Natural Resource and Bureau of Plant Industry permitting, compliance and reporting requirements associated with the propagation, planting and trading of Agarwood.

Board of Directors: Governance and Quality Oversight

MARCIANO GECOLEA
Agroforestry Director

Licensed Forester with a master's degree in resource management from Lincoln University, New Zealand and a BS Forestry degree from UP Los Baños

-highly experienced forestry professional with more than three decades of expertise, specializing in the complete agarwood production life cycle, among others. He possesses a strong foundation in sustainable agroforestry practices, land suitability evaluation, and GIS-driven and precision plantation management tools. His extensive knowledge of regulatory frameworks enables him to efficiently secure permits, oversee environmental impact assessments, and implement community-based livelihood initiatives that generate both ecological sustainability and economic growth.

As the former Regional Forestry Director for Asia Plantation Capital across Southeast Asia, Marciano successfully managed large-scale agarwood plantation projects—from strategic land acquisition and development to essential oil processing and value-added production.

His pioneering contributions to sustainable agarwood cultivation in the Philippines, combined with his background in watershed management and renewable energy initiatives, position him as a trusted expert. Marciano brings a distinctive blend of technical expertise, regulatory proficiency, and sustainable development leadership, making him an invaluable partner for investors seeking to establish responsible and profitable agarwood ventures.

R.A.G FERDINAND DOMINGO
Agri-Science Director

-Associate Professor and Director of PRMSU-Mango Research and Development Center, with a master's degree in Agricultural Technology Education from Central Luzon State University.

Over 40 years' experience with mango cultivation, SMART mango production and has conducted extensive research for the commercialization of pre and postproduction mango technologies.

Responsible for introduction of Dwarf Mango Technology and the promotion and development of integrated pest and disease management technologies.

Copyrighted research includes ecology and control of the mango Cecid fly, bio-ecology and control of mango twig borer and assessment of productivity enhancement measures for carabao mango.

Senior Management

ROMINA DALIT
Public Relations Manager

Over 12 years' experience in Government services, where she has held significant roles as secretary to the Mayor and Vice Governor. Here tenure in these positions has given her deep insights into public administration, leadership and Strategic Planning

Romina spent five years as a public relations manager, working for three major Philippine Corporations, honing her skills in managing corporate communications, fostering strong media relations and building the public image for prominent organisations.

A skilled Real Estate Broker with a strong background in land consolidation and expediting complex real estate transactions, title clearances and closure of property and land lease contracts.

Since 2021, she has worked in the capacity of a compliance officer for all related permitting requirements for Agarwood (import/export) CITIES, Bureau of Customs, Wildlife Cultural Permitting and Land Transportation Permits.

BILLY MEDEL
Plantation Manager

Billy launched his career with the Philippine National Oil Company-Energy Development Corporation (EDC), where he was responsible for all community relations as Community Partner (CP) and Professional Technician (PT). Established livelihood projects for establishing a variety of food-based crops.

In 2017 he was appointed Nursery Manager and was responsible for the importation and propagation of two million seedlings, before being promoted to plantation manager. His role also included land leasing and community relations.

Spent four years as Nursery and Plantation Manager for an essential oil processor, responsible for the cultivation and supply of Ylang Ylang, Patchouli, Vetiver grass and importation and propagation of Agarwood seedlings. Responsible for out-planting over 75,000 Agarwood saplings on behalf of third-party growers.

Institutional Collaborative Relationships

Golden Forests maintains collaborative relationships with leading Philippine agricultural research institutions and government agencies to support cultivation excellence, regulatory compliance, and continuous improvement.

President Ramon Magsaysay State University (PRMSU)

President Ramon Magsaysay State University (PRMSU) has established itself as a leader in mango research and development within the Province of Zambales. The university is recognised for founding the Regional Mango Centre, which focuses on scientific analysis, field trials, production advancements, and disease prevention technologies. Through its Regional Mango Centre, provides technical advisory support for Golden Forests' mango cultivation operations. The university's expertise in grafting techniques, dwarf variety development, and field trials directly supports the Company's proprietary Carabao-Elena hybrid development programme.

Key expertise provided:

- Scientific analysis and field trial design
- Grafting methodology and dwarf seedling development
- Disease prevention protocols and production advancement
- Quality control standards for commercial mango cultivation”

Visayas State University (VSU)

Visayas State University (VSU) plays a crucial role in research, training, and conservation regarding *Aquilaria* species, including *Aquilaria malaccensis*, a species prevalent in Leyte. VSU is notable for its research on rainforestation technology and for conducting studies focused on identifying ecological threats and developing effective propagation systems for *Aquilaria*. The university's staff contributes expertise in *Aquilaria* species research, rainforestation technology, and conservation genetics.

Key expertise provided:

- Research on the *Aquilaria* agarwood species and related species ecology
- Propagation system development and optimisation
- Identification of ecological threats and mitigation strategies
- Rain forestation integration methodologies

University of the Philippines Los Baños (UPLB) has different scientific people we work with

The University of the Philippines Los Baños (UPLB), particularly through its College of Forestry and Natural Resources (CFNR) and Institute of Biotechnology, is engaged in significant research programs centred on the conservation and sustainable production of agarwood-producing species, which support Golden Forest’s agarwood programme development. UPLB is at the forefront of developing advanced techniques such as DNA barcoding and innovative inoculation methods. These efforts support the preservation and responsible use of valuable plant resources. Currently, Golden Forests is participating in trials of a biofertilizer (MykoPlus), a product produced by UPLB, to assess the impact it will have on the natural production of agarwood resin. The Company

has also received proposals for developing optimized Inoculation Formulations for sustainable agarwood production in the Philippines.

Key expertise provided:

- DNA barcoding for *Aquilaria crassna* authentication
- Advanced inoculation methodology development
- Conservation genetics and sustainable production protocols
- Biotechnology applications in agroforestry

Department of Environment and Natural Resources (DENR)

Biodiversity Management Bureau (BMB): The BMB is responsible for regulating the propagation and trade of *Aquilaria* species in the Philippines. As these species are listed under CITES Appendix II and deemed endangered in the wild, the BMB plays a pivotal role in ensuring legal and sustainable practices. Golden Forests was the first Company to be certified to import *Aquilaria crassna* from Thailand into the Philippines. Golden Forests compliance requirements include the following permits and licenses:

- CITIES Import Permit
- Wildlife Cultural Permit
- Bureau of Plant Industry License to Operate
- BOC Certificate of Registration (import and export)

Department of Science and Technology

Philippine Council for Agriculture, Aquatic and Natural Resources Research and Development (DOST-PCAARRD). The DOST-PCAARRD monitors and funds research and development programs aimed at the conservation and sustainable production of agarwood in the Philippines. By supporting both scientific research and practical applications, this agency plays a vital part in promoting responsible resource management and innovation in the sector. DOST offer Golden Forests, in partnership with UPLB, a future source of funding for developing optimized Inoculation Formulations for sustainable agarwood production in the Philippines.

Scientific Advisory Support

Name	Expertise/Role	Key Contributions/Areas	Engagement Status
Dr Bong Salazar	Scientific expert in post harvesting mango technologies, pest and disease control.	Grower clustering approaches, introduced post harvesting technologies, protocols for pest and disease controls. Promotion of high-density mango farming	Advisor
Dr. Paciencia P. Milan	Mother of Rain forestation technology	Key figure in VSU initiatives, integration of *Aquilaria* into rain forestation farming systems	Informal
Mr. Manuelito R. Putong	Technical consultant at Aetherial Natural Oils Corp (ANOC), President of AGAP	Developing and implementing advanced, legal, and sustainable production methods	Informal

For. Rustum A. Aguilos	Project leader from ERDB-FWRDEC	Led research teams in scouting and assessing wild *Aquilaria* populations in Mindanao and the Visayas	Informal
Lorraine Cristy E. Ceniza and Marlito M. Bande	Researchers from Visayas State University	Published studies on conservation and ecological threats of *Aquilaria* on Leyte Island	Informal
Ramon Barba	Revolutionary figure	Discovered that spraying potassium nitrate could induce off-season flowering, enabling year-round production	Informal
Roel R. Rubion	Provided insights into PhilGAP for Mango	Explained requirements for certification and importance for global competitiveness	Informal
Enrico Batungbacal	Farm manager and owner of PhilGAP-certified farm	Demonstrated sustainable practices: fertigation, mechanized pruning, precision spraying	Informal
Mrs. Florida A. Domingo	Crop Science Coordinator	Pest management, productivity enhancement, mango variety research, improving mango production in Zambales and Central Luzon	Advisor

9. FINANCIAL PLAN AND PROJECTIONS

Overview

The financial plan covers a 15-year projection (2026-2040) for Golden Forests' agroforestry operations, covering *Aquilaria crassna* agarwood and Dwarf Sweet Elena Carabao Mango cultivation, alongside native species reforestation. The model demonstrates the Company's path to profitability, revenue scaling trajectory, and capital efficiency through its self-funding business model.

All projections are based on conservative assumptions regarding tree population growth, harvest yields, market pricing, and operational costs. Detailed financial modelling is provided in Appendix A, with supporting agronomic.

Revenue Projections

Agarwood Revenue Streams

Tree Sales (upfront capital):

- Initial pricing: USD 292.50 per tree (2026)
- Annual price escalation: 1.5% per annum
- Tree population growth: 10% per annum, commencing with 35,000 trees (2026)
- Volume discount application: 10% discount applied to gross sales to reflect volume purchase incentives

Oud Wood Chip Export Sales (commission revenue):

- Export sales commence: Year 9 (2034)
- Initial pricing: USD 775/kg premium-grade chips
- Annual price escalation: 1.5% per annum
- Company commission: 10% of gross chip sales
- Revenue recognition: Client receives 90% of gross proceeds; Company retains 10% commission
- Volume discount application: 10% discount applied to gross sales to reflect volume purchase incentives.

Bond Interest Income:

- Source: Prepaid expenses deducted from agarwood tree sales for future inoculation, harvesting, and extraction costs
- Interest rate: 4% per annum
- Purpose: Funds held in interest-bearing bonds until release for scheduled operational expenses (inoculation years 7-8, harvesting years 9-10, extraction years 9-10)

Mango Revenue Streams

Tree Sales (upfront capital):

- Initial pricing: USD 592.50 per tree (2026)
- Annual price escalation: 3% per annum
- Tree population growth: 10% per annum, commencing with 15,000 trees (2026)
- Volume discount application: 10% discount applied to gross sales to reflect volume purchase incentives

Mango Fruit Sales (commission revenue):

- Fruit sales commence: Year 5 (2030)
- Export pricing: USD 4.00/kg (70% of production)
- Domestic pricing: USD 3.00/kg (30% of production)
- Annual price escalation: 1.5% per annum
- Company commission: 20% of gross fruit sales
- Revenue recognition: Client receives 80% of gross proceeds; Company retains 20% commission
- Yield management: Projections based on 80% of planted trees reaching full fruiting capacity, providing 20% buffer for non-fruiting trees

Total Revenue Summary

Combined Revenue Projections (15 years):

- Total gross sales/revenue: USD 1.3 billion
- Net sales/revenue (after deductions): USD 574.9 million

Expense Projections

Sales and Marketing Costs:

- Agent commissions: Market-aligned commission structure for professional sales agents. Commission is paid on the gross sales price of the trees sold and a % share of the revenue the Company earns from the sale of Oud wood chips and mango fruits
- Marketing and sales budget: 5% of gross tree sales allocated to support digital marketing infrastructure, content creation, and brand development.
- Land lease owner and agent commissions: 1.75% of gross tree sales revenue and a % share of the revenue the Company earns from the sale of oud wood chips and mango fruits

Total projected Sales and Marketing Costs (15 years): USD 288.1 million

Cost of Sales/Services

Direct Materials:

- Agarwood and mango seedlings (DNA-verified/grafted cultivars)
- Native species (reforestation)
- Fertilisers, soil amendments, and agricultural inputs
- Inoculation materials and formulations (agarwood)
- Annual cost escalation: 3% per annum

Direct Labour:

- Nursery workers, plantation staff, harvest crews
- Costs scale with plantation expansion and operational activity
- Annual cost escalation: 6% per annum

Overhead:

- Security Personal and Caretakers
- Land lease payments
- Nursery and Staff house leases
- Nursery construction/expansion expenses
- Insurance premiums
- Depreciation on plantation infrastructure, equipment, and technology systems
- Transport and warehousing
- Annual cost escalation: 5% per annum

Total Projected Cost of Sales (15 years): USD 125.4 million

Operational Expenses:

- Management salaries and administrative expenses
- Security personnel and caretakers
- Land lease payments
- Insurance premiums
- Utilities, maintenance, and operational support
- Annual cost escalation: 5% per annum

Total projected Operational Expenses (15 years): USD 20.3 million

Total Projected Costs (15 years): USD 433.8 million

Pre-Paid Expenses

A distinctive feature of the Company's financial model is the prepaid expense structure for agarwood operations:

Mechanism:

- At the time of agarwood tree sale, costs for future inoculation, harvesting, and extraction services are deducted from gross tree sales revenue
- These funds are placed in interest-bearing bonds (4% per annum)
- Funds are released from bonds as expenses are incurred during the agarwood lifecycle

Expense Schedule:

- **Inoculation:** Years 7-8 (cost escalation 3% per annum)
- **Harvesting:** Years 9-10 (cost escalation 3% per annum)
- **Oud wood chip extraction:** Years 9-10 (cost escalation 4% per annum)

Accounting Treatment:

- Prepaid expenses are capitalised and held on the balance sheet
- Expenses are recognised over the 7-year establishment period before inoculation commences
- This treatment results in higher reported net income margins during early years, as major operational costs are deferred

Impact on Financial Metrics:

- The net income margin of 30.5% is partially attributable to this prepaid expense capitalisation
- Cash flow remains positive through bond interest income and staggered expense recognition
- This structure ensures capital availability for critical operational milestones whilst maintaining financial reporting transparency

Profitability Analysis

Gross Profit:

- Cumulative gross profit over the 15-year projection period: USD 444.9 million
- Gross profit margin: 44.2% of net sales
- Driver: Increasing productive tree volumes and commission revenue from maturing plantations

Net Income After Tax (NIAT):

- Cumulative NBIT (15 years): USD 429.1 million
- Cumulative NIAT (15 years): USD 321.85 million
- Net income margin: 30.5% of net sales
- Growth trajectory: Net income improves over time as operational scale increases and revenue diversification strengthens

Note on Net Income Margin: The high net income margin (30.5%) is significantly influenced by the capitalisation of agarwood pre-operational costs, which are held as prepaid expenses for an initial 7-year period before being recognised. This accounting treatment defers major expense recognition, resulting in higher reported profitability during establishment phases. Actual cash flow and operational profitability should be evaluated alongside reported net income to provide a complete financial picture.

Balance Sheet Highlights

Assets:

- Cash and cash equivalents: Growth from USD 3.3 million (2026) to USD 155.5 million (2040)
- Total assets: Increase from USD 11.2 million (2026) to USD 336.4 million (2040)
- Asset composition: Cash, accounts receivable, inventory (biological assets), property, plant and equipment, prepaid expenses (agarwood bonds)

Liabilities and Equity:

- Total liabilities: Decrease over projection period as loans are repaid and operational funding shifts to equity-based financing
- Total equity: Growth from a negative position in the initial year to USD 332 million (2040), driven by retained earnings and capital contributions
- Capital structure: Transition from debt-reliant early stage to equity-financed mature operations

Cash Flow Projections:

Net Cash Flow:

- Positive and growing throughout the projection period
- Annual net cash flow: Exceeding USD 24 million by year 15
- Cumulative strength: Demonstrates operational sustainability and self-funding capacity

Cash Ending Balance:

- Year 15 (2040): USD 155.5 million
- Indicator: Strong liquidity position supporting continued expansion, operational contingencies, and potential dividend distributions

Cash Flow Drivers:

- Upfront tree sales generating immediate capital inflows
- Staggered commission revenue as plantation matures
- Bond interest income supporting interim liquidity
- Disciplined cost management and operational efficiency gains

Key Financial Assumptions and Risk Factors

Revenue Assumptions:

- Tree population growth: 10% per annum (dependent on sales demand and capital availability)
- Tree pricing escalation: 1.5%-3% per annum (subject to inflation, competitive dynamics, and market conditions)
- Harvest yields: Based on conservative agronomic projections (*see Appendix C and Appendix D*)
- Market pricing: Mid-premium grade assumptions for agarwood; export-grade assumptions for mango
- Commission rates: 10% (agarwood), 20% (mango) - subject to competitive pressures and client negotiation

Cost Assumptions:

- Labour escalation: 6% per annum (subject to Philippine labour market conditions and wage inflation)
- Materials escalation: 3% per annum (subject to commodity pricing and supplier dynamics)
- Overhead escalation: 5% per annum (subject to general inflation and operational scaling)

Operational Assumptions:

- Mortality rates: 3-5% per annum, covered by 20% surplus planting buffer
- Mango fruiting success: 80% of planted trees reach full production (20% buffer)
- Operations escalation: 5-15% per annum (subject to general inflation and operational scaling.
- Processing, sales and delivery (oud wood chips/mango fruits): 1.5% of gross sales revenue
- Geographic concentration: All operations in Philippines (no geographic diversification in base case)
- Export market access: Continued compliance with CITES, phytosanitary, and import regulations

Tax and Regulatory Assumptions:

- Corporate income tax: 25% of net income (base assumption)
- Subject to approval: Philippine Board of Investments tax holiday - 5-year full waiver, followed by 10% rate for subsequent 10 years
- VAT: 12% on itemised expenses (materials, services, contract labour)
- Subject to approval: VAT waiver on importation of capital equipment

- Tax benefits are incorporated into projections but remain subject to regulatory approval and ongoing compliance

All assumptions are subject to the risks and uncertainties outlined in the Safe Harbour Statement and Section 11 (*Risk Assessment and Mitigation Strategies*).

Capital Expenditure (CapEx)

Plantation Infrastructure:

- Nursery infrastructure: Solar pumps, piping networks, misters, and sprinklers for temperature and irrigation management
- Irrigation systems: Solar pumps, piping networks, drip irrigation feeders installed per hectare
- Security infrastructure: Fencing, watchtowers, searchlights across plantation estates
- Agricultural technology: Drone fleet for pest/disease monitoring, aerial surveys, and precision application

CapEx Strategy:

- Front-loaded investment during plantation establishment phases
- Efficiency gains: Drone technology and precision systems reduce long-term labour costs
- Scalability: Infrastructure designed for modular expansion as plantation acreage increases

Sustainability and Impact Investments

Reforestation Programme:

- One native endemic species planted for each agarwood and mango tree sold
- Costs fully allocated as operating expenses (deducted from gross revenues)
- No direct revenue generation, but supports:
- Long-term environmental sustainability objectives
- Potential eligibility for carbon credits or ecosystem services programmes (under evaluation)
- Enhanced brand positioning and client value proposition

Mortality Guarantee Programme:

- Additional 20% surplus trees planted annually beyond client allocations
- All costs associated with surplus planting expensed as incurred
- No revenue projected from replacement tree harvests (conservative approach)
- Financial impact: Ensures client satisfaction and protects Company commission revenue continuity

Mango Yield Buffer:

- Annual mango yield projections based on 80% of total planted trees
- 20% buffer accounts for non-fruiting trees, disease impacts, or underperformance
- Conservative modelling approach reduces revenue volatility risk

Financial Model Sensitivity

Key Variables Impacting Projections:

High impact

- Tree sales volume (driven by sales and marketing effectiveness and client demand)
- Harvest market pricing (oud wood chips, mango fruit)
- Mortality rates exceeding buffer assumptions
- Mango fruiting trees exceeding the 20% buffer
- Regulatory approval of tax incentives

Moderate impact

- Labour cost escalation beyond 6% assumption
- Foreign exchange fluctuations (USD/PHP)
- Climate events affecting harvest timing or quality

Lower impact (within reasonable ranges)

- Materials cost escalation (covered by pricing escalation assumptions)
- Bond interest rate variations
- Overhead cost variations

Scenario analysis:

Investors should consider financial outcomes under multiple scenarios, including:

- **Base case:** Assumptions as presented
- **Conservative case:** Lower tree sales growth (e.g., 5% vs 10%), higher mortality (e.g., 7% vs 3-5%), lower harvest pricing (e.g., -10% from base)
- **Optimistic case:** Higher tree sales growth (e.g., 15%), lower mortality (e.g., 2%), premium pricing achievement (e.g., +15% from base)

The financial forecast (*see Appendix A*) is fully integrated and provides for detailed scenario modelling. Conservative and Optimistic cases are available upon request.

Strategic Financial Observations

Self-funding growth model: The financial projections demonstrate Golden Forests' ability to finance expansion through operational cash flow rather than continuous external capital raises. Tree sales generate immediate capital, whilst deferred commission revenue creates recurring income as plantations mature. This model supports disciplined scaling with limited equity dilution.

Revenue diversification: The dual-crop model (agarwood and mango) with different maturation timelines creates staggered revenue streams:

- Mango commission revenue: Commencing year 5, continuing 20+ years
- Agarwood commission revenue: Commencing year 9, concentrated in years 9-10

This diversification reduces cash flow volatility and operational risk.

High-margin recurring revenue: Commission revenue requires minimal incremental cost once plantations are established, supporting strong EBITDA margins (44.2% gross profit margin) as operations scale. This high-margin profile becomes increasingly significant as mature plantation acreage grows.

Capital efficiency: The prepaid expense/bond structure for agarwood operations enables the Company to collect capital upfront whilst deferring major operational costs, creating favourable working capital dynamics and reducing reliance on debt financing during establishment phases.

Path to profitability: The model demonstrates a clear path to sustained profitability with cumulative net income (NAIT) of USD 321.8 million over 15 years, strong cash generation (USD 155.5 million ending cash balance), and a healthy equity position (USD 322 million by year 15).

Appendix references

- **Appendix A:** Complete 15-year financial model including detailed income statements, balance sheets, cash flow statements, and supporting schedules
- **Appendix B:** Calculation tool for determining the projected annual return on investment for agarwood and mango trees
- **Appendix C:** Third Party Validation of the yield of oud wood chips from Aquilaria Agarwood trees
- **Appendix D:** Lifecycle progression data, and productivity benchmarks
- **Appendix E**: Domestic and Export Pricing Trends

All forward-looking financial statements and projections are subject to the assumptions, risks, and qualifications outlined in the Safe Harbour Statement and Section 11 (Risk Assessment and Mitigation Strategies).

10. INVESTMENT HIGHLIGHTS

Investment Thesis

Golden Forests represents a compelling investment opportunity in the high-growth sustainable agriculture sector, combining proven agroforestry cultivation expertise with a scalable, capital-efficient business model. The Company addresses rising global demand for premium agarwood and mango products through professionally managed plantation operations, delivering attractive returns to both tree-purchasing clients and Company equity investors.

The investment case is underpinned by:

- Proven market demand for premium agarwood and Carabao Mango in established export markets
- Self-funding growth model reducing reliance on continuous external capital
- High-margin recurring revenue from a commission-based business structure
- Geographic and crop scalability enabling multiple expansion pathways
- Strong financial projections demonstrating a clear path to profitability and cash generation
- Experienced management with deep operational and scientific expertise
- Regulatory positioning, including CITES accreditation and planned forest certifications

Market Opportunity

Agarwood market:

- Global market valuation: USD 41+ billion (*Persistence Market Research 2024*), projected to reach USD 84.9 billion by 2034 [*Precedence Research 2024*]
- Supply-constrained industry due to wild population depletion and CITES regulation
- Plantation-cultivated agarwood is increasingly preferred for traceability and sustainability
- Premium oud oil pricing: Up to USD 40,000 per litre for top-quality grades (*ScienceDirect 2020-2021*)
- Strong demand drivers: Middle East religious/cultural use, luxury fragrance sector growth, natural wellness trends

Mango market:

- Global market: USD 55.20 billion in 2025 and is projected to reach USD 68.30 billion by 2030 (*Moore Intelligence 2025*).
- Philippine mango export value: Estimated at USD 1.6 billion annually (*Mordor Intelligence 2025, Research and Markets 2025, Worldmetrics 2025*).
- Carabao variety commands premium pricing in key markets (Japan, South Korea, the US, Middle East)
- Quality differentiation: Superior Brix levels (18-24°) vs competing varieties (13-16°), and are ranked number one worldwide for sweetness (*JournalismReport.com 2025*)
- Year-round production capability: Golden Forests' proprietary dwarf Sweet Elena Carabao Mango and induced flowering techniques enable continuous harvest vs seasonal competitors

Agarwood Operations

Regulatory Leadership:

- CITES-accredited permit holder for *Aquilaria crassna* import/export in the Philippines positions Golden Forests as a compliant, traceable supplier in the regulated market
- Planned FSC (Forest Stewardship Council) certification application scheduled for Q4 2027, strengthening premium market access
- Competitive moat: Many agarwood producers lack proper certification, limiting export market access

Proprietary Technology:

- Exclusive licensing of patented and proven inoculation technology
- Option to acquire the inoculation technology copyrights for private label branding and direct sales in the Philippines
- Inoculation formulas and techniques specifically optimised for *Aquilaria crassna*
- Demonstrated efficacy in maximising both oud yield and resin quality
- Competitive advantage: Inoculation expertise directly impacts product grade, pricing, and client returns

Species Selection:

- *Aquilaria crassna* chosen for market preference, consistent resin formation, and quality control advantages
- DNA-verified seedlings ensure species authenticity and supply chain traceability
- Detailed species comparison vs *Aquilaria malaccensis* provided in Appendix C

Diversified Plantation Strategy:

- Multiple plantation locations in Southern Zambales optimise yield potential and reduce geographic concentration risk
- Site selection based on rigorous soil, climate, and topography assessment protocols
- **Expansion strategy:** Proven model replicable across Southeast Asia and South Asia

Mango Operations

Scientific Partnership Excellence:

- Collaboration with President Ramon Magsaysay State University Mango Institute ensures cultivation best practices and continuous improvement
- Access to leading Philippine mango research, grafting expertise, and disease prevention protocols
- **Scientific credibility:** Partnership validates operational standards and quality commitment

Elite Variety Development:

- Proprietary Dwarf Sweet Elena Carabao hybrid developed by Magsaysay State University scientists
- Dwarfing techniques enable high-density planting (416 trees per hectare vs traditional 100-150)

- **Unique characteristics:** Superior flavour, enhanced environmental resilience, strong market appeal
- **Year-round production:** Induced flowering techniques enable continuous flowering vs seasonal competitors, reducing revenue volatility

Operational Efficiency:

- High-density farming maximises land productivity and revenue per hectare
- AI-driven agricultural technologies optimise resource use and reduce labour costs
- Precision farming: Drones, smart sensors, and predictive analytics support consistent quality and yield

Certification Pathway:

- Planned PEFC (Programme for the Endorsement of Forest Certification) certification by 2027 (*see Appendix H*)
- Demonstrates commitment to responsible environmental stewardship and sustainable practices
- **Market advantage:** Certification increasingly required for premium export channels

Technology and Operational Excellence

Agroforestry Intelligence Platform

Proprietary AI-enabled monitoring and management system integrating:

- Real-time soil condition tracking (moisture, nutrients, pH)
- Climate and micro-weather analysis for irrigation optimisation
- Early pest and disease detection through pattern recognition
- Growth tracking and yield forecasting via predictive analytics
- Harvest timing optimisation based on maturity indicators

Operational Benefits:

- Precision resource allocation reduces water consumption and chemical inputs
- Early intervention capabilities minimise crop losses
- Data-driven decision-making improves consistency and quality
- Scalable technology infrastructure supports geographic expansion without proportional overhead increases

Drone-Supported Operations:

- Aerial surveillance for large-scale crop health assessment
- Targeted input application (fertilisers, pesticides), reducing waste and environmental impact
- Yield estimation and harvest planning support
- **Cost efficiency:** Technology reduces manual labour requirements whilst improving outcomes

Risk Mitigation and Client Satisfaction

Mortality Protection Programme:

- 100% tree replacement guarantee for specified adverse events (lightning, typhoons, drought, floods, pests, diseases, animal damage, theft, vandalism)
- 20% surplus planting buffer substantially exceeds historical mortality rates (3-5%)
- **Client confidence:** Guarantee protects client holdings and Company commission revenue continuity

Quality Assurance Framework:

- GPS-verified tree allocation with transparent traceability
- Annual independent auditing and reporting
- Client portal access for ongoing monitoring
- Plantation visit programme enabling direct asset verification

Conservative Financial Modelling:

- Yield assumptions based on proven cultivation methodologies, not optimistic projections
- Mango projections assume 80% fruiting success (20% buffer)
- Gross tree sales revenue reflects a 10% discount off the retail price
- Mortality buffers exceed historical experience
- Pricing assumptions based on mid-premium grades, not super-premium outliers

These risk mitigation measures protect both client outcomes and Company financial performance, reducing operational volatility and supporting sustainable growth.

Financial Highlights and Returns

Company Financial Performance (15-Year Projections):

- Total projected revenue: USD 1.3 billion (gross), USD 574.9 million (net)
- Cumulative net income after tax: USD 321.8 million
- Net income margin: 30.5%
- Gross profit margin: 44.2%
- Ending cash position (Year 15): USD 155.5 million
- Total equity (Year 15): USD 322 million

Client Return Profile (Evidence of Business Model Strength):

- Agarwood clients: Projected 18% annualised returns over 10-year cycle
- Mango clients: Projected 12% annualized returns (years 1-15), increasing to 20%+ (years 16-25)

Strong client returns drive demand, enable premium pricing, support organic growth through referrals, and validate the Company's operational excellence and market positioning.

Self-Funding Growth Characteristics:

- Upfront tree sales generate immediate capital for plantation expansion
- Deferred commission revenue creates recurring cash flow without incremental capital requirements
- Model supports 10% annual growth without continuous external fundraising
- Capital efficiency reduces equity dilution and enhances investor returns

Environmental and Social Impact

Sustainability Commitment:

- **Reforestation programme:** One native endemic species planted for every agarwood and mango tree sold
- **Target:** Restoration of denuded lands in Southern Zambales through systematic native species planting
- **Species diversity:** Narra, Agoho, Mangono, Molave, Talisay, Belite
- **Impact:** Ecosystem restoration, biodiversity enhancement, carbon sequestration

Community Development:

- Local employment generation through plantation operations
- Skills development and agricultural training programmes
- Economic contribution to rural Zambales communities
- Partnership with local landowners through lease arrangements

Potential Carbon Credit Opportunities:

- Reforestation activities may qualify for carbon offset programmes (under evaluation)
- Additional revenue stream possibility whilst delivering measurable environmental benefit
- Alignment with corporate ESG and impact investment mandates

Governance and Expertise

Multi-Jurisdictional Structure:

- ADGM (Abu Dhabi): Group governance and capital management
- Dubai: International sales and marketing operations
- Philippines (CADI): Plantation operations and asset management
- Structure provides: Clear operational separation, regulatory compliance, tax efficiency, market access optimisation

Scientific Advisory Panel:

- Leading experts across mango and agarwood cultivation disciplines

Institutional Collaborative Relationships with:

- President Ramon Magsaysay State University (mango excellence)
- Visayas State University (*Aquilaria* research)
- University of the Philippines Los Baños (biotechnology and conservation genetics)
- Philippine Department of Environment and Natural Resources (regulatory compliance)

- Department of Science and Technology - PCAARRD (R&D support)

Operational Track Record:

- 80+ years combined plantation management experience
- Proven cultivation methodologies validated through scientific partnerships
- Technology-enabled precision farming reduces operational risk
- Established export relationships in target markets

Detailed management team profiles are provided in Section 8 (*Company Organisation*)

Growth Strategy and Expansion Pathways

Near-Term Focus (Years 1-5):

- Establish operational excellence in Philippine plantations
- Achieve FSC (agarwood) and PEFC (mango) certifications
- Build a client base across the Middle East, Europe, and Southeast Asia markets
- Commence harvest operations (mango year 5, agarwood year 9)
- Demonstrate consistent quality and client satisfaction

Medium-Term Expansion (Years 5-10):

- Geographic diversification: Evaluate suitable sites
- Crop diversification: Assess additional high-value agroforestry species (subject to market analysis and scientific validation)
- Value-added processing: Develop mango by-product commercialisation (dried fruit, seed oil, cosmetic ingredients)
- Scale client acquisition: Expand agent network, strengthen digital marketing infrastructure
- Potential capital markets activity: Evaluate funding options for accelerated growth

Long-Term Vision (Years 10+):

- Multi-country operations across Southeast and South Asia
- Diversified crop portfolio capturing premium agroforestry market segments
- Integrated processing for producing agarwood essential oils and value-added product capabilities
- Established brand recognition as a premium sustainable agriculture platform
- Potential liquidity events for equity investors (trade sale, PE exit, public markets)

Investment Risks

Prospective investors should carefully review Section 11 (*Risk Assessment and Mitigation Strategies*) and the Safe Harbour Statement, which outline material risks including:

- Biological risks (crop disease, pest outbreaks, mortality events)
- Environmental risks (climate variability, extreme weather, natural disasters)
- Market risks (commodity pricing volatility, demand fluctuations, competitive pressures)

- Operational risks (execution challenges, labour availability, technology performance)
- Regulatory risks (CITES compliance, export restrictions, tax incentive approval)
- Financial risks (foreign exchange, interest rates, capital availability)
- Geographic concentration (current operations solely in the Philippines)

Golden Forests has implemented comprehensive risk mitigation strategies, but investors should understand that agriculture investments carry inherent uncertainties and that actual results may differ materially from projections.

Investment Opportunity Summary

Golden Forests offers investors exposure to:

- **High-growth market sectors** with demonstrated global demand and supply constraints
- **Proven business model** generating strong client returns and Company profitability
- **Capital-efficient structure** enabling self-funded expansion with limited dilution
- **Technology-enabled operations** provide competitive advantages and scalability
- **Experienced management** with deep scientific partnerships and operational expertise
- **Regulatory positioning** supporting premium market access and export capabilities
- **Multiple expansion pathways** across geographies and crop types
- **Sustainability impact** delivering measurable environmental and social benefits
- **Attractive financial projections** with a clear path to profitability and cash generation

The combination of market opportunity, operational excellence, financial strength, and expansion optionality positions Golden Forests as a compelling investment in the sustainable agriculture sector.

Investment Terms: Senior Secured Loan

Term	Description
Principal Amount	$350,000
Interest Rate	8% per annum (fixed)
Loan Tenure	12 months (bullet repayment at maturity)
Collateral Assets	Vested Biological Assets: • 600 Agarwood trees • 300 Carabao mango trees
Collateral Value	Agarwood: $175,500 (retail value) Mango: $177,750 (retail value)
Security Structure	Senior secured loan with registered security interest over trees
Additional Security	10% equity shares held in escrow, convertible if loan not repaid after 12 months
Mortality Guarantee	100% replacement of trees at no cost to lender if any die
Projected Returns	Agarwood: 10-year lifespan, 18% avg. annual return Mango: 25-year lifespan, 12% annual return for 15 years, 20% for next 10 years
Reporting	Annual Audited report on tree health, yield, and replacement activity
Use of Proceeds	Marketing and sales program, Land leasing, Working capital Repayment of short-term loans

11. RISK ASSESSMENT AND MITIGATION STRATEGIES

Overview

Golden Forests' agroforestry operations are subject to environmental, operational, market, regulatory, and natural hazard risks inherent to agricultural businesses in tropical regions. The Company has implemented comprehensive risk mitigation protocols across all material risk categories to protect both operational performance and client outcomes.

Regulatory and Compliance Risks

CITES Compliance (*Aquilaria* species):

- *Risk:* Failure to maintain CITES permits or documentation could prevent agarwood exports

Mitigation: Maintain rigorous compliance protocols; employ dedicated compliance staff; conduct regular audits; maintain strong relationships with DENR-BMB; ensure complete traceability from planting through export

Tax Incentive Approval:

- *Risk:* Philippine Board of Investments tax incentives (5-year holiday, subsequent 10% rate) may not be approved
- *Mitigation:* Financial model includes base case with standard 25% tax rate; incentives treated as upside, not base assumption; pursue approval through proper channels with professional advisory support

Export Market Access:

- *Risk:* Changes in import regulations, phytosanitary requirements, or trade relationships could restrict market access
- *Mitigation:* Maintain certifications (FSC, PEFC); diversify across multiple export markets; monitor regulatory developments; build relationships with importers and trade associations

Operational Risks

Inoculation Performance (Agarwood):

- *Risk:* Inoculation may not produce expected resin quality or quantity, impacting harvest value
- *Mitigation:* Utilize proven inoculation technology licensed from Forestry Industry Organisation (FIO) of Thailand; apply protocols developed through extensive field trials; monitor resin development post-inoculation; maintain conservative yield assumptions

Pest and Disease Outbreaks:

- *Risk:* Unexpected pest or disease pressure could cause significant crop losses beyond buffer assumptions
- *Mitigation:* AI-enabled early detection systems; regular monitoring and scouting; IPM protocols; rapid response capabilities; maintain 20% mortality buffer; insurance coverage where available

Technology Performance:

- *Risk:* AI systems, sensors, or drones may malfunction or underperform, reducing operational efficiency
- *Mitigation:* Redundant systems for critical functions; regular maintenance and calibration; backup manual processes; vendor support agreements; staff training on both automated and manual methods

Supply Chain Disruptions:

- *Risk:* Disruptions in seedling supply, agricultural inputs, or export logistics could delay operations
- *Mitigation:* Diversify supplier relationships; maintain strategic inventory buffers; develop contingency logistics plans; vertical integration of nursery operations

Financial Risks

Foreign Exchange Volatility:

- *Risk:* PHP/USD exchange rate fluctuations impact tree pricing, costs, and harvest proceeds
- *Mitigation:* Pricing in USD where feasible; natural hedge through USD-denominated revenues and PHP-denominated costs; consider hedging strategies for large transactions

Client Demand Variability:

- *Risk:* Tree sales may not achieve the projected 10% annual growth, impacting expansion plans
- *Mitigation:* Diversified marketing across geographies and client segments; agent network expansion; digital marketing optimisation; strong client value proposition (18-20% returns); conservative growth assumptions

Capital Availability:

- *Risk:* Inability to raise additional capital if needed for accelerated growth or unforeseen costs
- *Mitigation:* Self-funding model reduces capital dependency; strong cash generation from operations; maintains relationships with potential capital providers; conservative financial planning

Geographic Concentration Risk

Single-Country Operations:

- *Risk:* All current operations in the Philippines create concentration risk (political, regulatory, and natural disasters)
- *Mitigation:* The Philippines was chosen for its favourable conditions and regulatory environment; proven track record in mango export; expansion strategy includes geographic diversification to Thailand, Vietnam, and Cambodia over the medium term; concentration acceptable during proof-of-concept phase

Market and Competitive Risks

Agarwood Market Dynamics:

- *Risk:* Oversupply from new plantations, synthetic alternatives, or demand shifts could depress pricing
- *Mitigation:* Premium quality focus (DNA-verified genetics, proven inoculation); certification (CITES, FSC) creating differentiation; long cultivation cycle limits supply response; strong demand fundamentals in the Middle East and Asia

Mango Market Competition:

- *Risk:* Competition from Mexico, India, Thailand, and other major exporters could pressure pricing
- *Mitigation:* Superior Carabao variety quality (higher Brix, better flavour); year-round production capability vs seasonal competitors; certification (PEFC, PhilGAP); established Philippine mango brand recognition; focus on premium markets (Japan, South Korea)

Risk Management Framework

Ongoing Monitoring:

- Regular risk assessments and mitigation plan updates
- Key risk indicators (KRIs) tracked across environmental, operational, financial, and market categories
- Quarterly risk review with the management team
- Annual comprehensive risk assessment with Board oversight

Insurance Coverage:

- Crop insurance is available and economically justified
- Property and casualty insurance for infrastructure and equipment
- Business interruption insurance
- Directors' and officers' liability insurance
- Professional indemnity for technical services

Contingency Planning:

- Business continuity plans for natural disasters, disease outbreaks, and operational disruptions
- Alternative supplier and logistics arrangements
- Emergency response protocols for environmental incidents
- Financial reserves for unforeseen circumstances

Stakeholder Communication:

- Transparent risk disclosure to clients and investors
- Regular reporting on risk events and mitigation actions
- Clear escalation procedures for material risk developments
- Open channels for stakeholder concerns and feedback

Appendix F identifies key risk categories, assesses site-specific considerations for Southern Zambales operations, and outlines mitigation measures designed to minimise the probability and impact of adverse events.

Conclusion

Golden Forests acknowledges that agroforestry operations inherently involve biological, environmental, market, and operational risks. The Company has implemented comprehensive risk identification, assessment, and mitigation protocols designed to minimise adverse impacts whilst maintaining operational excellence and client satisfaction.

However, investors and clients should understand that despite robust mitigation measures, actual outcomes may differ materially from projections due to factors beyond the Company's control. All financial projections, yield assumptions, and return expectations are subject to the risks outlined in this section and the qualifications set forth in the Safe Harbour Statement.

The Company remains committed to continuous improvement in risk management practices, proactive adaptation to changing conditions, and transparent communication regarding material risk developments.

APPENDIX A

15-year financial forecast: 2026-2040

(Excel File attached separately)

APPENDIX B

Calculation tool for determining the projected annual return on investment for agarwood and mango trees

(Excel File attached separately)

APPENDIX C

Third-party validation of the yield of oud wood chips from *Aquilaria crassna* agarwood trees harvested in years 9 and 10 (2 years post-inoculation)

Source / Institutional Author	Projected Yield (kg per tree)	Reference / Publication Title	Web Address / Digital Identifier
Karnataka Forest Department	3.0 – 4.0	Cultivation of Agarwood (Official Technical Bulletin)	aranya.gov.in
B. Kanwal et al. (Frontiers in Plant Science)	1.5 – 2.0	Induction of Agarwood: A Review of Artificial Methods	doi.org/10.3389/fpls.2023.1114061
ScienceDirect / Journal of Cleaner Production	1.2 – 2.5	Sustainable Agarwood Production in SE Asia	sciencedirect.com
Tropical Plantations Ltd (TPL)	2.5 – 3.0	Operational Progress Report: *A. crassna*	tropicalplantations.com
Asia-Pacific Forestry Commission (FAO)	1.5 – 2.0	Non-Wood Forest Products: Agarwood	fao.org/forestry
ResearchGate / University of Malaysia	1.8 – 2.2	Efficiency of Fungal Inoculants in *A. crassna*	researchgate.net

APPENDIX D

Lifecycle progression data and productivity benchmarks of agarwood and mango trees

Agarwood (*Aquilaria crassna)*

Stage / Year	Key Activity / Milestone	Yield / Output (kg per tree)	Productivity Notes
Year 0	Seedling propagation & nursery	-	DNA-verified seedlings, nursery phase lasts 8 months
Year 1–2	Out-planting & intercropping (cassava)	-	Cassava provides shade; removed after 2 years
Year 1–5	Establishment & maintenance	-	Regular weeding, pruning, fertilisation, irrigation, pest/disease management
Year 5	Trees reach 7–10m height, 10–15cm dbh*	-	Suitable for inoculation
Year 7–8	Inoculation phase	-	Proprietary fungal/bacterial inoculation to induce resin formation
Year 9	First harvest (oud wood chips)	~1.5	Oud wood chip yield per tree is based on conservative, validated methodologies
Year 10	Second/final harvest (oud wood chips)	~2.0	Oud wood chip yield per tree, based on conservative, validated methodologies
Year 9–10	Oud chip extraction & export	Oud wood chips	premium grade chips targeted for export and downstream processing of essential oils
Buffer/Surplus	Mortality buffer	20% surplus planted	Ensures replacement for losses due to natural causes
Reference Yields	Third-party validation	1.5–2.0 (FAO, Kanwal)	See Appendix G for institutional benchmarks (Karnataka, FAO, University of Malaysia, etc.)

*DBH (diameter at breast height), measured at 1.3 metres above ground level, is used to classify tree size and growth stage within the plantation.

Mango (Dwarf Sweet Elena carabao)

Stage / Year	Key Activity / Milestone	Yield / Output (kg per tree)	Productivity Notes
Year 0	Grafting & nursery	-	Carabao-Elena hybrid, dwarf rootstock for high-density planting (416 trees/ha)
Year 1–4	Establishment & growth	-	Regular care, irrigation, pest/disease management
Year 5	First fruiting	~30	Initial yield: trees begin annual fruit production
Year 6	Yield ramp-up	~50	Output increases each year
Year 7	Yield ramp-up	~70	Output increases each year
Year 8	Yield ramp-up	~85	Output increases each year
Year 9	Yield ramp-up	~95	Output increases each year

Year 10–25	Peak/stable production	~100 per year	Stable annual yield per tree, year-round harvest enabled by flower induction
Year 5–25	Harvest & export	70% for export	Premium fruit, strict grading for export markets
Year 5-25	Harvest & Domestic Sales	30% for domestic consumption	High quality fruit, inspected and distributed to domestic markets
Buffer/Surplus	Non-fruiting buffer	20% of trees as buffer	Projections based on only 80% of trees reaching full fruiting capacity
Value-Added	By-products	Dried, juice, pulp, etc.	Additional revenue streams under evaluation

APPENDIX E

Export and retail pricing trends for agarwood oud wood chips and Carabao Mango fruits

Export pricing trends for premium mangoes from the Philippines

Historical export prices (2017–2021)

Between 2017 and 2021, export prices for Philippine mangoes ranged from USD 3.28 to USD 3.87 per kg. In recent years, there has been a slight decrease in average export prices.

Export price overview (2023–2024)

During 2023, Philippine mangoes, primarily the Carabao variety, were exported at prices ranging from USD 2.20 to USD 5.51 per kg. In 2024, export prices exhibited notable volatility, expanding to a range from USD 1.83 to USD 8.86 per kg. This variation reflects the dynamic nature of the international mango market.

Price forecasts (2026-2030)

Most sources anticipate a CAGR of 4–5% for export prices through 2030, with premium Carabao Mangos commanding the highest prices. By 2030, export prices for premium Philippine mangos are expected to reach a minimum of USD 4.00–USD 6.00 per kg, depending on grade, destination, and market conditions. In high-value markets (e.g., Japan, Australia), retail prices can be much higher due to logistics, tariffs, and premium branding.

Key insights on export pricing

Retail prices for premium mangoes are highest in urban centres and during lean months, with substantial regional and seasonal differences. Export prices fluctuate depending on quality, destination, and global supply-demand dynamics. Carabao Mangoes, recognised as a premium variety, command higher prices in international markets such as Japan, Korea, and the Middle East.

Retail price trends for premium mangoes in the Philippines

The retail price for premium mangos in the Philippines (including Carabao Mango) currently ranges between USD 3.45 and USD 8.09 per kg (PHP 191.78 to PHP 450.31 per kg) in major cities like Manila and Quezon. Wholesale prices are lower, between USD 2.41 and USD 5.66 per kg.

National retail price range (January 2026)

Retail prices for mangoes, including Carabao, range from USD 3.45 to USD 8.09 per kg in Manila and Quezon. The average retail price is USD 3.45 per kg, with a peak price of USD 4.65 per kg recorded in October 2024, and a low price of USD 2.18 per kg observed in June 2024. As of April 15, 2025, the retail price stands at USD 2.87 per kg.

Market outlook

The retail sector in the Philippines is projected to grow at 7–8% CAGR through 2030, with food and beverage (including mangos) benefiting from resilient domestic consumption and rising disposable incomes.

Premium mangos, especially Carabao, are expected to maintain their price premium due to quality, export demand, and branding.

Summary table

Retail price forecast for premium mangos (2026–2030)

Year	Estimated Retail Price Range (USD/kg)	Estimated Retail Price Range (PHP/kg)
2026	3.45–8.09	191–450
2027	3.50–8.30	195–460
2028	3.55–8.55	200–470
2029	3.60–8.80	205–480
2030	3.65–9.00	210–500

Note: Prices may vary by region, season, and export demand. Annual escalation is based on historical trends and market forecasts.

Key insights on retail pricing

Retail prices for premium mangoes in the Philippines are highest in urban centres and during lean months, with significant regional and seasonal variation. Export prices are influenced by quality, destination, and global supply-demand dynamics. Premium Carabao Mangoes fetch higher prices in international markets, particularly in Japan, Korea, and the Middle East.

Summary

The quoted prices of USD 3.00 per kg (domestic) and USD 4.00 per kg (export) for mango fruit are within the typical range for premium or export-quality mangoes, especially for high-value varieties like Carabao Mango from the Philippines. Price variations are influenced by country, quality, and market destination.

Key sources for premium mango price trends

- National Mango Board (USDA Market News Service)
- Tridge – Global Wholesale Market Prices
- IMARC Group – Mangoes Pricing Report
- Selina Wamucii – Mangoes Price Insights
- Expana – Mango Prices and Global Trade Data
- Procurement Resource – Mango Price Trend
- USDA Economic Research Service (ERS) – Fruit and Vegetable Prices
- GEPA Exporters Portal – Global Market Overview of Mangoes

Export trends for *Aquilaria Crassna* Oud Wood Chips

Current export pricing (2025–2026)

Low grade

Vietnamese plantation oud wood chips with low resin content are priced at USD 1.45–USD 2.30 per g (USD 1,450–USD 2,300 per kg) for bulk orders of 500 g or more. In Thailand, bulk export prices are approximately USD 600-USD 800 per kg, with price variations depending on supplier and grade.

Medium grade

Vietnamese cultivated oud wood chips with medium resin content and AA+ to AAA+ grading typically range from USD 110–USD 160 per 100 g (USD 1,100–USD 1,600 per kg) for AA+ grade, and USD 129–USD 150 per 100 g (USD 1,290–USD 1,500 per kg) for 3A+ grade. Thailand bulk prices remain at USD 600-USD 800 per kg for medium-grade chips.

High grade

Premium, high-resin oud wood chips, including wild or top plantation varieties, command higher prices. Vietnam Ky Nam (premium) oud wood chips are priced at USD 2,500–USD 3,000 per kg for high resin, premium quality. Other high-grade listings are available at USD 1,320–USD 1,550 per kg for chips with high oil and resin content.

Prices are influenced by resin content, chip size, origin (Vietnam, Thailand, Cambodia), and certification requirements such as CITES and phytosanitary standards.

Future export pricing trends and market outlook (2026 and beyond)

The global agarwood oud wood chip market, including *Aquilaria crassna*, is projected to grow at a CAGR of 6.4–7.4% from 2024 to 2033. Market value is expected to rise from approximately USD 10.7 billion in 2025 to USD 17.6 billion by 2033.

Price drivers include rising demand in luxury perfumery, incense, and wellness sectors, especially in the Middle East and Southeast Asia. Sustainability and certification, such as CITES, are increasingly important, favouring plantation-grown and certified chips. Market prices may fluctuate due to supply chain disruptions, regulatory changes, and shifts in consumer demand.

Summary

The quoted price of USD 775 per kg for mid-premium grade agarwood oud wood chips is at the lower end of the global price range and aligns with prices for plantation-grown, certified chips in Southeast Asia, including the Philippines.

Key references and sources

- Alibaba: *Aquilaria crassna* oud wood chips – price listings
- Global Trade Plaza: Agar-wood chips (*Aquilaria Crassna*)

- Fatma Perfumes: Agarwood Price Guide 2025
- Persistence Market Research: Agarwood Chips Market Size & Trends 2024–2031
- Straits Research: Agarwood Chip Market Size, Share & Growth Report by 2033

APPENDIX F

Risk assessment and mitigating strategies: Southern Zambales, Philippines

Category		Risk/Impact	Site-Specific Considerations (Southern Zambales)	Mitigation Measures
Environmental	**Water Resources**	Depletion of local water sources due to high irrigation demand, especially during the dry season, leading to water scarcity for plantation operations.	Proximity to coastal areas increases the risk of saltwater intrusion into freshwater aquifers if groundwater is over-extracted.	Implement efficient irrigation (e.g., drip irrigation) managed by AI-enabled monitoring. Prioritise rainwater harvesting and storage infrastructure. Explore alternative, sustainable water sources. Conduct regular groundwater levels and salinity monitoring. Establish water use efficiency targets and tracking.
	Water & Soil Quality	Contamination of soil and waterways from chemical pesticides and fertilisers via agricultural runoff, impacting ecosystem health and product quality.	Potential for existing heavy metal contamination in parts of Zambales to be exacerbated by agricultural chemical use.	Employ Integrated Pest Management (IPM) protocols minimizing chemical dependency. Utilise precision farming practices engaging AI-driven application systems. Conduct regular soil and water quality monitoring and testing. Establish buffer zones around water bodies. Use organic and low-impact agricultural inputs where feasible
	Biodiversity	Habitat loss and reduced local biodiversity from converting native vegetation to large-scale, monoculture plantations.	The area has a balance of rural/mountainous areas and coastal ecosystems (mangroves, marine life) requiring protection.	Intercrop *Aquilaria* with mangoes and other native species. Avoid clearing primary forests or critical habitats. I Implement 1:1 native species reforestation programme.

				Maintain wildlife corridors and habitat connectivity. Partner with conservation organisations for biodiversity monitoring.
	Soil Degradation	Soil erosion and sedimentation of rivers and irrigation canals, especially in sloped terrain or areas with vegetation removal.	Intense rainfall during typhoons can accelerate soil erosion, a major concern given the region's climate vulnerability.	Implement soil conservation measures: contour farming, terracing, and mulching. Maintain vegetative cover to stabilise soil. Establish erosion control structures in vulnerable areas. Monito soil health indicators and implement remediation as needed. Use cover crops and intercropping to protect soil surface.
Socio economic	**Livelihoods & Land Use**	Potential land use conflicts and displacement of local communities due to land conversion for plantations. Competition with existing economic activities.	Sothern Zambales is balancing fishing, tourism/resort development, and other agriculture; land for plantations may compete with these established uses.	Engage in transparent community consultations before land acquisition. Formalise land rights and ensure fair lease agreements with landholders. Respect indigenous peoples' rights and traditional land use. Provide training and alternative livelihood opportunities. Support community development initiatives.
	Market & Finance	Unstable commodity prices for agarwood and mango products, high investment costs; lengthy time horizons before revenue generation; foreign exchange volatility.	Mango prices can be volatile due to pest issues, and quality defects.	Integrate precision farming technology (drones, sensors) for early pests/diseases detection. Implement strict quality control protocols to maintain premium positioning. Diversity across two crops (agarwood and mangoes) with different maturing timelines.

				Hedge foreign exchange exposure where practical. Maintain conservative financial assumptions and buffers.
	Labour	Aging farmer population and declining youth interest in agriculture, leading to potential labour shortages and skills gap.	Philippine agricultural sector faces labour challenges common across developing economies.	Offer competitive wages, performance incentives and career advancement opportunities. Provide modern, technology-focused training to attract younger workers. Create clear pathways from field positions to supervisory and management roles. Invest in employee development and retention programmes. Utilise technology to reduce labour intensity where feasible.
Natural hazards and climate change	**Volcanic Activity**	Lahars (volcanic mudflows) and severe flooding from Mount Pinatubo's volcanic deposits during heavy rainfall events.	Southern Zambales is downstream of the Pinatubo region; although further south than most-affected areas (Botolan), still vulnerable to regional flooding from Lahar flows.	Establish effective early warning systems and evacuation protocols. Collaborate with government on river management and infrastructure maintenance (dikes/dredging). Site plantations outside highest-risk lahar flow paths Maintain insurance coverage for catastrophic events Note: No volcanic activity in Southern Zambales for the past 400 years, reducing eruption risk.
	Earthquakes	Structural damage to plantation infrastructure and potential liquefaction or landslides from seismic activity.	The region faces moderate earthquake risk due to its proximity to the Philippine Fault System.	Design and construct all critical infrastructure to earthquake-resilient standards. Conduct geohazard assessments before site development.

				Implement structural reinforcement for processing and storage facilities. Maintain business continuity and disaster recovery plans. Ensure adequate insurance coverage.
	Climate Change	Reduced yields due to extreme weather (prolonged droughts, intense rainfall events, heatwaves); shifting climate patterns disrupting traditional growing seasons; increased pest and disease incidence.	Coastal location increases vulnerability to sea-level rise and storm surges affecting low-lying plantation areas.	Install AI-managed drip irrigation for precise water delivery and drought resilience. Use climate-resilient crop varieties and proven varieties and proven genetics. Implement water management strategies for both *El Niño* (drought) and *La Niña* (flooding0 scenarios. I Install shade structures for young *Aquilaria* trees during establishment. Apply mycorrhiza inoculants to enhance drought tolerance and nutrient uptake. Monitor climate trends and adapt cultivation practices proactively.

All forward-looking statements and financial projections are subject to these risks and uncertainties, as detailed in the Safe Harbour Statement.

APPENDIX G

Comparative analysis: *Aquilaria crassna* and *Aquilaria malaccensis* agarwood species

Topic	*Aquilaria crassna*	*Aquilaria malaccensis*
Demand	• The USA is emerging as the largest growth market for agarwood oud wood chips. • China is the largest market for Agarwood essential oils (primarily for medicinal purposes and religious ceremonies • Other key export markets for agarwood oud wood chips are UAE, Saudia Arabia, Japan, Singapore, and Taiwan. • In India, agarwood is in high demand for traditional Ayurvedic remedies. • UK and Germany are the top European markets for pure agarwood oil.	
Quality	• *Aquilaria crassna* produced from Vietnam is the highest quality for supplying the fragrance industry.	• The scent from *Aquilaria Malaccensis* is less woody and sweeter than that of *Aquilaria crassna*
Location	• *Aquilaria crassna* is not grown in the Philippines because it is an exotic species. A DENR Administrative Order dated Feb 10th, 2025 allowed for the ownership of exotic species for use on privately owned lands. • Vietnam and Thailand are major suppliers of *Aquilaria crassna* for the past 20 years. Laos, Cambodia, and Malaysia cultivate this species but on a small scale.	• *Aquilaria Malaccensis* is endemic to the Philippines, but it is an endangered species as there are few managed plantations and the wild trees are over 10 years old and subject to poaching. • *Aquilaria Malaccensis* cultivated in India, Malaysia, Laos, and Cambodia are from large managed plantations.
Authentication	• *Aquilaria crassna* trees are known for its generous size of the tree and elliptic, leathery, smooth leaves. It exhibits Pear-shaped fruit obovoid and huge (3-4 cm) *Aquilaria crassna* does not exist legally in the Philippines as of Feb 1st, 2025. • Thailand maintains an online national agarwood registration system for control and monitoring of *Aquilaria crassna* plantations.	• There are 11 species of *Aquilaria* found in the Philippines and without DNA tests it is hard to determine between the species. *Aquilaria Malaccensis is the only native species of value in the Philippinrs.* • All other species of *Aquilaria* in the Philippines are of no value.
Inoculatoin	• The inoculation formulae for inducing stress through fungi injections for *Aquilaria crassna* is commercially proven and applied in year 7 after out-planting.	• Recently the University of Malaysia has released an inoculation technology for applying to *Aquilaria Malaccensis*

		with ongoing trials for induction in year 5.
MARKET SHARE	• *Aquilaria crassna* currently holds the largest global market share (21.25%) and is on an upward growth trend (*ScienceDirect.com*)	• *Aquilaria malaccensis* has a global market share of 20.23% of which India supplies the largest volume and commands a lower price point due to the quality (*ScienceDirect.com*).
SUPPLY CHANNELS	• *Aquilaria crassna* can is imported from Thailand where there are only a 4-hour transit time and subsequently minimal stress to the seedlings. Due to the high number of Mother Stock trees, high volumes of *Aquilaria crassna* seedlings are available for export between June-January each year. All agarwood exported from Thailand is plantation origin.	• There are a limited number of *Aquilaria Malaccensis* mother stock trees located in the Philippines that retain a wildlife cultural permit allowing for trade. Most mother stock trees are in the western provinces (Leyte, Mindanao, Sorsogon, Samar. • Importation from countries such as India results in long transit time and increased stress and mortalities as a result.
Oil processing and extraction	• Hydro distillation produces the highest quality of oil, however, is labour intensive. • Pressurised steam is faster but produces a lower quality. • There are currently new techniques being prototyped	• Currently, there are no processing facilities for extracting agarwood oil in the Philippines
Bi-products	• *Aquilaria crassna* leaves form a key ingredient in traditional medicine for bone fractures, bruises, high blood pressure, constipation, headaches, diabetes, digestive ailments, and mild sedation.	• Both *Aquilaria crassna* and *Aquilaria Malaccensis* produce by products after the oil extraction. The remaining wood still retains aromatic qualities and reduced to powder (exhausted powder) for use in incense or bakhoor (scented bricks). The powder pressed into small statues, is often of a religious significance. Lower grades of wood produce carved objects like sculptures and beads that retain their aromatic qualities for years, although it appears that the beads in circulation are of other woods soaked in agarwood oil.

Unique aroma	● *Aquilaria crassna* is rich in sources of sesquiterpenes, responsible for the agarwood's unique aroma and therapeutic properties. The main sesquiterpenes found in *Aquilaria Crassna* include (A) agarospiranes, (B) eudesmanes, (C) eremophilanes, (D) guaianes, (E) humulenes and (F) other sesquiterpenoids. There are only 5 other species that have these sesquiterpene properties.	● *Aquilaria Malaccensis* has these unique sesquiterpene properties, however, are less rich in the source of sesquiterpenes.

APPENDIX H

PEFC ACCREDITATION: requirements and processes for mango plantations

Introduction to PEFC certification for mango plantations

The Program for the Endorsement of Forest Certification (PEFC) is a voluntary, third-party certification system designed to verify that plantations are managed according to sustainable practices. While PEFC has traditionally focused on timber products, recent developments have expanded its scope to include "Trees Outside Forests" (TOF), such as fruit plantations. Notably, in 2023, the first mango plantation in Asia, located in Thailand, achieved PEFC certification, underscoring the applicability of these standards to fruit crops like mangoes.

Overview and purpose of PEFC certification

PEFC certification for mangoes ensures that production methods conserve biodiversity, support local communities, maintain economic viability, and respect the rights of workers. There are two main types of PEFC certification relevant to mango plantations:

Sustainable Forest Management (SFM): This certification assesses the management practices on the plantation itself, ensuring alignment with sustainable and responsible land stewardship.

Chain of Custody (CoC): This certification tracks mangoes from the plantation through the supply chain to the consumer, guaranteeing traceability and integrity throughout all stages.

Accreditation and requirements in the Philippines (2026)

The Philippines is currently implementing the Philippine Sustainable Forest Certification System (PSFCS) to harmonise local practices with PEFC international standards. Below are the key requirements and processes for securing accreditation in the country:

1. Eligibility and Standards

Land status: Plantations must meet the national definition of a "forest" or "Trees Outside Forests," which typically involves minimum area, tree crown cover, and height criteria.

Management plan: Plantation owners are required to maintain a documented management plan that addresses environmental protection (including biodiversity and soil health), social responsibility, and economic stability.

Legal compliance: All operations must adhere to local laws, such as the Revised Forestry Code (PD 705) and regulations set forth by the Department of Environment and Natural Resources (DENR).

2. Process for affiliation

Preparation: Align farm practices with the guidelines established under the PSFCS.

Select a certification body: Engage a PEFC-recognised, third-party auditor operating in the Philippines to conduct an independent assessment.

Audit: The auditor performs field visits, documentation reviews, and interviews to verify compliance with national SFM standards.

Issuance: Upon meeting compliance requirements, a certificate is issued, generally valid for five years and subject to annual surveillance audits.

3. Additional national requirements

In addition to PEFC requirements, mango growers in the Philippines may need to fulfil specific local accreditations:

Mango classifiers: Individuals handling mangoes must obtain accreditation from the National Accreditation Committee (NAC) through the Department of Agriculture (DA).

Good Agricultural Practices (GAP): Many mango farms, especially those targeting export markets, first secure PhilGAP certification as a foundation for food safety and sustainability.

Regulatory agencies and their roles

The DA is the primary government agency responsible for regulating and certifying mango plantations in the Philippines. Several bureaus and programs within the DA have distinct roles:

Bureau of Plant Industry (BPI):

Nursery accreditation: Through the Crop Production Division (BPI-CPD), BPI accredits plant nurseries to ensure high-quality planting materials.

Pest and disease management: BPI oversees pest surveillance, provides early warning information, and enforces quarantine laws (e.g., for Mango Pulp Weevil).

Phytosanitary certification: The National Plant Quarantine Services Division (NPQSD) is the sole agency authorised to issue Phytosanitary Certificates required for mango exports.

Bureau of Agriculture and Fisheries Standards (BAFS):

Formulates and enforces Good Agricultural Practices (GAP) certificates for fruit and vegetable farms and accredits "Mango Classifiers" to uphold national quality standards.

High Value Crops Development Program (HVCDP): Coordinates the overall development of the mango industry, including the implementation of the Philippine Mango Industry Roadmap.

Department of Trade and Industry (DTI):

Assists with Geographical Indication (GI) certification and export promotion, complementing the DA's role in production.

Transition to high-density planting

In the Philippines, shifting from traditional large-tree systems to high-density planting is a pivotal approach for achieving Good Agricultural Practices (GAP) compliance. High-density planting, characterised by smaller trees, enables safer and more efficient orchard management.

PEFC certification for *Aquilaria crassna* plantations

The Program for the Endorsement of Forest Certification (PEFC) provides third-party verification for the sustainable management of *Aquilaria crassna* (agarwood) plantations. In the Philippines, this certification is integrated into the Philippine Sustainable Forest Certification System (PSFCS), ensuring that forest products from these plantations receive international market recognition for responsible and sustainable practices.

Certification requirements for *Aquilaria crassna* plantations (2026)

Due to the highly regulated and endangered status of *Aquilaria crassna* under CITES, certification in the Philippines involves strict legal and environmental standards:

Legal land tenure: Applicants must submit a certified photocopy of their land title, which may be the Original Land Title, Transfer Certificate of Title (TCT), Certificate of Land Ownership Award (CLOA), or, for untitled Alienable and Disposable (A&D) lands, a Tax Declaration.

Wildlife culture permit (WCuP): This permit is mandatory for any commercial activity involving the cultivation, harvesting, or trade of Aquilaria species. It certifies that the material is sourced from plantation-grown trees and not from the wild.

Seed/seedling traceability: Documentation is required to prove that seeds or seedlings were legally sourced. This typically involves submitting a CITES Import Permit for exotic species and a Local Transport Permit for moving planting materials within the country.

Management plan: A comprehensive forest management plan must be prepared, reviewed by a Registered Professional Forester, and approved by the Department of Environment and Natural Resources (DENR). The plan should address biodiversity protection, socio-economic benefits, and sustainable harvesting guidelines.

Accreditation and affiliation process

Register the Plantation: Apply for tree plantation registration to the local Community Environment and Natural Resources Office (CENRO).

Obtain WCuP: Apply for a Wildlife Culture Permit at the DENR regional office.

Engage a certification body: Choose a PEFC-accredited third-party auditor in the Philippines, typically those trained under the PSFCS framework, to conduct an independent assessment of the plantation.

Field audit: The auditor performs an evaluation based on PSFCS standards, which include criteria for environmental impact, labour rights, and legal compliance.

Chain of Custody (CoC): If the plantation is involved in processing agarwood products (such as oil or chips), obtaining a CoC certificate is necessary. This certificate ensures that the product is traceable from the certified plantation through to the final consumer.

Key regulatory bodies

DENR-Forest Management Bureau (FMB): Responsible for the development and management of the national forest certification system.

Biodiversity Management Bureau (BMB): Oversees the issuance of CITES permits and regulates wildlife culture for endangered species, including *Aquilaria crassna*.

Philippine Sustainable Forest Certification System (PSFCS): Acts as the national governing body, endorsed by PEFC, to manage local forest certifications.

APPENDIX I

Detailed assessment of durian for future cultivation and diversification

Introduction

Golden Forests is exploring the feasibility of integrating durian plantations into its existing agroforestry operations. This document provides a comprehensive overview of durian cultivation in the Philippines, drawing upon established guidelines, best practices, and regional development plans. It aims to inform the Company's strategic decision-making process by outlining key considerations for successful durian plantation development, including regulatory compliance, optimal growing conditions, and effective management practices. This plan considers the potential for durian to enhance Golden Forests' portfolio, aligning with sustainable agriculture principles and contributing to both local and export markets.

Markets and popular durian cultivars/varieties for Thailand and the Philippines

Durian is a highly valued fruit in Southeast Asia, with Thailand and the Philippines being prominent producers offering diverse cultivars that vary in aroma, flavour, texture, and market appeal. This section provides an overview of the most popular durian varieties from both countries, emphasising their key sensory characteristics, consumer preferences, and suitability for domestic consumption and export markets. Understanding these cultivars is essential for strategic production planning and positioning in competitive international markets, particularly given the growing demand from countries such as China. The following tables summarise notable varieties, highlighting those favoured for their mild or distinctive aromas, quality, and commercial viability:

Category	Cultivar / Variety	Origin	Aroma & Flavour Profile	Texture & Other Notes	Market & Export Notes
Thailand: mild-aroma varieties	**Monthong (Golden Pillow)**	Thailand	Mild aroma, sweet with caramel & vanilla hints	Thick, creamy, smooth, meaty flesh	Most common Thai variety; top export choice; beginner-friendly
	Kanyao (Long Stem)	Thailand	Balanced sweet & floral aroma	Soft, creamy texture	Premium variety; long ripeness window; mild scent
	Gradum-tong	Thailand	Mildly sweet, less creamy aroma	Bright yellow, soft flesh; can be mushy if overripe	Less common; softer texture
	Chanthaburi No. 1 & No. 3	Thailand	Virtually odourless (No.1) or odourless for 3 days post-harvest	Suitable for odourless transport	Government-developed hybrids for export
	Specialty Regional Varieties	Thailand	Floral or mild aromas (Long Laplae, Lin Laplae, PaLa-U)	Varied textures; milder scents	Regional niche markets
	Duyaya	Thailand/Philippines	Mildest flavour, very light on pungency	Light texture	Also grown in the Philippines; noted for mildness

Philippines: popular varieties	**Puyat**	Hybrid (Chanee x Native)	Sweet-bitter, thick yellow-orange flesh	Pear-shaped large spikes; farmer favourite	Most widely grown; primary export variety
	Arancillo	Hybrid (Chanee x Native)	Creamy, subtle aroma	Pale yellow flesh; milder taste	Top commercial choice; quality export
	UPLB Gold	Philippines	Sweet, pest-resistant	Thick skin; good edible portion	Locally developed; emerging commercial variety
	Musang King (Mao Shan Wang)	Malaysia (also PH)	Buttery, bittersweet, premium flavour	Creamy texture	High demand premium export variety
	Chanee	Thailand	Bolder, intense aroma	Parent variety of Puyat, Arancillo	Introduced hybrid
	Kanyao	Thailand	Mild, floral aroma	Long peduncle	Also grown locally
	Mamer (DES916)	Philippines	Highly rated native variety	—	Local specialty
	Duyaya	Philippines	Sweet, mild aroma	Large fruit; high flesh-to-seed ratio	Less common but notable
	Alcon Fancy	Philippines	Mild flavour/aroma	Bright yellow, thick flesh	Rare specialty
	Graviolens (Guintung)	Philippines	Unique cheesy/avocado-like taste	Small, colourful flesh	Native, less commercial
	Bisaya, Umali	Philippines	Unique local flavour profiles	Varied	Native local cultivars
Export focus (Philippines, 2026)	**Puyat**	Philippines	Sweet, creamy, thick flesh	Preferred for export, especially to China	Main export cultivar
	Duyaya	Philippines	Sweet, large fruit	Growing premium export interest	Showcased in international expos
	D101	Malaysia/PH	Buttery, intense sweetness	Popular Malaysian cultivar grown locally	Exported to meet premium demand
	Arancillo	Philippines	Creamy, aromatic	Exported in smaller volumes	High-quality export variety

This table offers a clear, comparative overview of durian varieties relevant for market positioning and agroforestry planning.

Notes:

- **Thailand** leads with mild-aroma cultivars like *Monthong* and *Kanyao,* ideal for new consumers and export, alongside odourless hybrids for logistics.

- **Philippine durians** are dominated by hybrids derived from Thai varieties (*Puyat, Arancillo*), local specialities (*Mamer, Bisaya*), and premium imports (*Musang King*).
- Export markets, particularly China, favour *Puyat*, *Duyaya*, and *D101* for their flavour and export suitability.

Durian variant suitability, yield and estimated revenue performance

Below is a summary table of the three most recommended durian varieties for Philippine cultivation, based on site suitability, average yield per tree and per hectare, and estimated revenue per tree and hectare. The data are organised by the three main Philippine regions.

Region	Variety	Site Suitability	Average Yield (kg/tree)	Estimated Yield (kg/ha)	Estimated Revenue per Tree (USD)	Estimated Revenue per Hectare (USD)
Mindanao	Puyat	Well-adapted to local climate and soil	15–20	1,500–2,000 (100 trees/ha)	30–40	30,000–40,000
	Arancillo	Thrives in Mindanao conditions	14–18	1,400–1,800	28–36	28,000–36,000
	Musang King	Suitable for premium niche markets	16–22	1,600–2,200	32–44	32,000–44,000
Luzon	UPLB Gold	Pest-resistant; adapted to Luzon's environment	12–16	1,200–1,600	24–32	24,000–32,000
	Duyaya	Performs well under Luzon conditions	13–17	1,300–1,700	26–34	26,000–34,000
	Monthong	Grows under managed conditions; imported variety	14–18	1,400–1,800	28–36	28,000–36,000
Visayas	Puyat	Adaptable to Visayas' climate	14–18	1,400–1,800	28–36	28,000–36,000
	Arancillo	Performs well with good management	13–17	1,300–1,700	26–34	26,000–34,000
	Alcon Fancy	Suitable for niche sites with specific conditions	10–14	1,000–1,400	$0–28	20,000–28,000

This table provides a practical overview for selecting durian varieties based on agronomic suitability, productivity, and economic returns across the main Philippine regions.

Notes:

- Yield estimates assume an average planting density of 100 trees per hectare.
- Revenue per tree and hectare is calculated using an average price of approximately USD 2 per kg.
- Actual yields and revenues may vary depending on site management, climatic conditions, and market fluctuations.

Conclusion and Recommendations

Durian cultivation in Southeast Asia, particularly in Thailand and the Philippines, presents significant opportunities due to the diverse range of cultivars with varying aroma, flavour, texture, and market appeal. Thailand's mild-aroma varieties like Monthong and Kanyao lead in export markets and consumer acceptance, especially among new Durian consumers. Meanwhile, the Philippines offers a mix of hybrids derived from Thai varieties (such as Puyat and Arancillo),

native cultivars, and premium imports like Musang King, catering to both domestic consumption and expanding export demand, notably in China.

For strategic cultivation in the Philippines, the choice of durian varieties should align with regional site suitability, yield potential, and profitability. Based on current agronomic and market data, the following recommendations are made:

- **Mindanao:** Puyat, Arancillo, and Musang King are highly suitable, combining strong adaptability with high yields and substantial revenue potential.
- **Luzon:** UPLB Gold, Duyaya, and imported Monthong offer pest resistance, good performance, and market acceptance under local conditions.
- **Visayas:** Puyat and Arancillo remain reliable choices, supported by the niche-market potential of Alcon Fancy.

These selections balance agronomic factors with economic returns, facilitating sustainable and profitable Durian production across the Philippines. Growers and stakeholders are encouraged to consider these varieties within integrated production plans to maximise market opportunities while ensuring adaptability to local growing environments.

Key export markets of Philippine durian

1. **China:** By far the most dominant market, accounting for **95%** of global durian imports. The Philippines secured fresh durian market access to China in early 2023.
2. **Hong Kong & Singapore:** Traditional steady markets for fresh Philippine durian.
3. **Japan & South Korea:** Emerging markets with growing interest in both fresh and frozen durian products.
4. **Other Destinations:** The Philippines also exports to Thailand (primarily frozen for processing), Malaysia, Saudi Arabia, and the United Arab Emirates.

Rate of Increase

- **China market growth:** In 2024, Philippine durian exports to China reached USD 32.5 million, a staggering 144.4% **increase** from the previous year.
- **Sales milestones:** During the 2025 China-ASEAN Expo (CAEXPO), fresh Durian enquiries and interest led to more than USD 18 million in export sales generated by participating Philippine enterprises in a single event.
- **Volume surge:** Total exports reached 9,351 metric tonnes in the first nine months of 2024, more than doubling the total volume for all of 2023.
- **Future outlook:** The global durian market is projected to grow at a CAGR of 6.5% to 7% through 2033.

Global export/import of durian:

As of 2026, the global Durian market continues to experience a significant surge, primarily driven by China's immense consumption and a competitive supply landscape in Southeast Asia.

1. Global supply share (by country)

Thailand remains the dominant global producer, although its share is being challenged by Vietnam's rapid expansion.

- **Thailand:** ~57% to 65% of global exports (down from >90% in 2022).
- **Vietnam:** ~41.5% of the Chinese market share (up significantly since 2022).
- **Malaysia:** ~1.44% to 3% of global volume; focuses on high-value premium varieties like Musang King.
- **Philippines:** ~0.28% of the global market; rapid growth since gaining China market access in 2023.

2. Global demand (by region)

Demand is overwhelmingly concentrated in Asia, with China acting as the primary engine for market growth.

- **China:** Accounts for 95% of worldwide Durian consumption. In 2024, China imported a record USD 6.99 billion worth of Durian.
- **Asia-Pacific (Excluding China:** Accounts for approximately 85% of regional market share when production is included, with Singapore (3%) being another notable importer.
- **Western markets:** North America (4%) and Europe (7%) are experiencing a "steady uptick" in demand, driven by the Asian diaspora and new culinary interest.

3. Market value and growth projections

The market is shifting from a niche speciality to a major global commodity.

- **Total market valuation:** The global durian fruit market is valued at approximately USD 29.09 billion in 2025.
- **Growth rate:** Projected to reach USD 48.15 billion by 2033, with a CAGR of 6.5% to 7.5%.
- **Philippine performance:** Philippine exports to China reached USD 32.5 million in 2024, a 144.4% year-on-year increase.

4. Summary of supply-demand balance

Metric	2024-2025 status	2026-2033 outlook
Supply leader	Thailand (High Volume)	Vietnam & Philippines (Gaining share)
Demand centre	China (95% of total)	China remains dominant; India/Middle East emerging
Key export variety	Monthong (Thailand)	Musang King (Malaysia), Puyat (Philippines)
Value trend	High/unsaturated prices	Potential price softening as supply diversifies

Financial plan (work-in-progress)

Developing a durian plantation requires a robust financial plan based on specific assumptions about costs, revenue, and market conditions in the Philippines. Standard operating procedures have been drafted for the Philippines.

Required financial assumptions

These assumptions are critical for calculating startup costs, operating expenses, profitability, and return on investment (ROI).

- Startup capital & investment:
  - **Initial investment:** The total upfront capital needed for land preparation, seedlings, labour, infrastructure (irrigation, fences, storage), and initial maintenance (Years 1-3).

- **Source of capital:** Assumption regarding how the investment will be funded (e.g., bank loans, investors, etc.).

- Cost assumptions:
  - **Land cost:** The price of lease and/or acquisition of land per hectare varies significantly between Mindanao, Luzon, and Visayas.
  - **Seedling cost:** Price per high-quality, grafted seedling of commercial varieties like 'Puyat' or 'Arancillo'.
  - **Labour costs:** Daily or monthly wages for skilled and unskilled labour (planting, maintenance, harvesting, security).
  - **Material costs:** Prices of fertilisers (which can fluctuate), pesticides, tools, and farm equipment.
  - **Operational expenses:** Assumed annual maintenance costs, including irrigation, fertilisation, pest control, and security once the trees mature (e.g., around **PHP 6 million annually** for a 50-hectare plantation).
  - **Depreciation:** The straight-line method is often used to depreciate the value of farm tools and machinery over their estimated economic life.

- Revenue & market assumptions:
  - **Farm-gate price:** The expected selling price per kg of durian sold directly from the farm. This has recently increased from PHP 20-25 to PHP 50-70 per kg due to export demand.
  - **Yield projections:** Expected yield per tree and per hectare as the plantation matures (e.g., a 50-hectare plantation at age eight can yield around 550,000 kg annually).
  - **Maturity timeline:** Assumption on when trees begin commercial fruiting (typically Years 4-5 for grafted varieties) and reach peak production (Year 8+).
  - **Market share:** Percentage of produce designated for local markets versus high-value export markets (e.g., China, which pays premium prices).

- General economic assumptions:
  - **Inflation rate:** The projected annual inflation rate in the Philippines (forecasted between 2-4% through 2028).
  - **Exchange rate:** The anticipated peso-dollar exchange rate (pegged at PHP 56-58 for 2025-2028), which impacts export earnings and imported inputs.

Optional financial assumptions

- **Financing terms:** Specific interest rates and loan tenor if seeking external financing.
- **Market volatility buffer:** Assumption of price fluctuations due to seasonal gluts or competition from Thailand/Vietnam.
- **Climate/calamity impact:** Assumed potential losses due to weather-related disruptions (typhoons, drought), especially for sites outside of Mindanao's typhoon belt.
- **Processing revenue:** Assumption of revenue from value-added processing (e.g., frozen durian meat, paste) if post-harvest facilities are included in the plan.
- **Carbon credits:** Potential income generation from sustainable practices or rainforestation projects.

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APPENDIX J

Overview of durian Cultivation in the Philippines

Durian production in the Philippines is a collaborative effort between government agencies and industry organisations. Key players include:

Government agencies:

- **Department of Agriculture (DA):** The primary agency responsible for the industry's growth.
- **High Value Crops Development Program (HVCDP):** Manages the "Durian Protocol" to boost production, distributing high-quality planting materials and expanding cultivation areas.
- **Bureau of Plant Industry (BPI):** Responsible for crop research, production support, and certifying farms for export.
- **Department of Science and Technology - PCAARRD:** Leads the Science and Technology (S&T) programme for durian, focusing on increasing fruit yield, improving quality, and extending harvesting seasons through research and development.
- **Mindanao Development Authority (MinDA):** Coordinates regional development strategies and international partnerships to scale up durian production, specifically in Mindanao.
- **Department of Trade and Industry (DTI):** Supports producers through market promotion, product development, and innovation initiatives like the "One Town, One Product" (OTOP) programme.
- **Non-Government Organisations** (NGOs) and Associations:
  - **Durian Industry Association of Davao City (DIADC):** Organises major industry events and provides training on good agricultural practices (GAP).
  - **CAMP Asia:** A South Korea-based NGO that partners with local institutions to establish sustainable agricultural environments for Durian producer organisations in Mindanao.
- **Durian Exporters Association of the Philippines:** Focuses on strengthening the global competitiveness of Filipino durian and addressing logistical and production challenges for export.

Regulatory framework and standards

The Department of Agriculture (DA) maintains national standards and specific protocols for durian, primarily focused on quality grading and phytosanitary requirements for export.

Philippine National Standard (PNS):

1. PNS/BAFS 12:2024 (Durian – Product Standard – Grading and Classification): Provides the national criteria for commercial fresh durian, covering quality and classification, presentation, safety, and nutritional composition.

2. "Durian Protocol" (export protocols):

- **Refers to the protocol of phytosanitary requirements**: Bilateral agreements between the Philippines and importing countries.
- **China Protocol:** Signed in 2023, mandates that all exported durian must come from DA-registered farms and packing houses that adhere to strict pest management and sanitation rules.
- **Phytosanitary certification:** The Bureau of Plant Industry (BPI) strictly enforces these protocols, requiring exporters to obtain a Phytosanitary Certificate to prove the fruit is pest-free before shipment.

3. Production and farming guidelines:

- **Durian production guide:** A comprehensive manual outlining best practices for land preparation, spacing (e.g., 8m x 8m), and fertilisation cycles.
- **PhilGAP (Philippine Good Agricultural Practices):** The DA encourages and assists farmers in achieving PhilGAP certification, which is often a prerequisite for participating in the national export protocols.

4. Ideal locations and environmental requirements for the development of durian plantations:

- **Primary Hub:** Davao region remains the premier location.
- **Emerging Upland Areas:** Regions between 600-800 metres above sea level in Bukidnon and Cotabato.
- **Elevation Constraints:** Avoid areas above 800–900 metres.
- **Critical environmental requirements:**
  - **Soil:** Deep, well-drained sandy loam or clay loam rich in organic matter. Ideal soil pH is 5.5 to 6.5.
  - **Climate:** Hot and humid with temperatures between 24°C and 32°C.
  - **Water:** At least 1,500 mm of annual rainfall is needed. A short dry spell of 1–2 months is necessary to trigger and synchronise flowering.
  - **Sunlight:** Minimum of six hours of direct sunlight daily for optimal fruit production.

5. Key cultural management practices

- **Spacing and density:**
  - Traditional: 10m x 10m (100 trees/ha) or 8m x 8m.
  - Modern high-density: 6m x 6m to 7m x 7m, requiring intensive pruning to manage canopy size.
- **Planting protocol:** Use Large Planting Materials (LPMs) maintained in a nursery for over a year for better survival. Plant at the start of the rainy season and provide temporary shade (e.g., coconut leaves or bananas) for at least three months to prevent transplant shock.
- **Fertilisation:** Transition from nitrogen-rich fertilisers for young trees to potassium-heavy blends during fruit development to enhance sweetness. Apply fertilisers monthly for immature trees and quarterly for bearing trees.
- **Irrigation:** Modern plantations increasingly use drip irrigation delivery, which is critical during fruit set when roots are sensitive to water stress.
- **Integrated Pest Management (IPM):** Essential for export-standard fruit, focusing on controlling Phytophthora (root rot) and fruit borers through regular pruning for air circulation and strategic fungicide/insecticide use.

Durian plantation development timelines in the Philippines

Period	**Mindanao (Core Region)**	**Luzon (Emerging)**	**Visayas (Diversifying)**
Pre-2000s	Established as the national "Durian republic".	Scattered backyard trees in Bicol and Southern Tagalog.	Limited small-scale planting in Eastern Visayas (Leyte/Samar).
2000-2015	Shift to commercial clones (Puyat, Arancillo).	Technical trials in Cagayan Valley to assess climate suitability.	Early commercial interest: DA distributes seedlings to diversify coconut-based farms.
2017-2022	Export boom (1780% value growth).	Emerging "off-season" production identified in parts of Luzon.	Small-scale commercialisation in Negros; focus on local tourism demand.

2023	China Protocol signed; first direct air-shipments.	Market access expansion: Luzon farms begin seeking GAP/certification.	Evaluation of typhoon-resilient areas for large-scale cluster farming.
2024	Export volumes double; 15,000-ha expansion begins.	Adoption of updated national quality standards (PNS/BAFS 12:2024).	DA-HVCDP intensifies "Climate-Smart" durian training for Leyte/Samar growers.
2025	First frozen durian shipments to China; focus on "Class A" quality.	Increase in grafted seedling distribution to northern highland borders.	Integration of durian into regional "High-Value" roadmaps for Western Visayas.
2027	IoT-driven irrigation and high-density planting are standardised.	Commercial luxury harvests for Metro Manila markets.	Focus on consolidation Hubs to link Visayan smallholders to export supply chains.

Development considerations by region (2027 Context)

- **Mindanao:** Plans focus on cluster farming to consolidate smallholders for bulk international export. Priority is placed on regions like Davao City, Caraga, and Zamboanga Sibugay.
- **Luzon:** Focus on proximity to Metro Manila markets and strategic timing to hit high-price windows.
- **Visayas:** Plans prioritise climate-smart agriculture, focusing on areas with volcanic soil (like Negros) or protected valleys to mitigate typhoon risks.

Durian plantation development work programme (2027)

Phase	Activity	Mindanao Focus	Luzon/Visayas Focus	Timeline
I. Pre-Development & Site Suitability	Soil & climate assessment (pH 5.5-6.5; 1,500mm+ rain).	High-density expansion in Davao/Caraga.	Identification of typhoon-resilient microclimates.	Month 1-2
II. Technical Design & Road mapping	Selection of commercial clones (Puyat, Duyaya, Arancillo).	Integration with export cluster hubs.	Pilot testing of IoT systems for off-season growth.	Month 2-3
III. Capacity Building & Training	PhilGAP (Good Agricultural Practices) orientation.	Specialised training on export-grade (Class A) harvesting.	Technical training via DA School-on-the-Air (SOA).	Month 3-4
IV. Land Preparation & Infrastructure	Clearing, ploughing, and staking (8mx 8m or 6mx 6m).	Installation of modern drip irrigation systems.	Establishment of nursery sheds for large planting materials (LPM).	Month 4-6
V. Planting & Initial Maintenance	Seedling transplanting at the start of the rainy season.	Large-scale "Durian 500" tribal community planting.	Diversified planting with intercrops (e.g., banana, coconut).	Month 6-
VI. Monitoring & Certification	Enrolment in DA-BPI farm registration for export.	Monitoring of IoT/S&T interventions for yield.	Finalising local market linkages and OTOP branding.	Ongoing

Summary

Integrating durian plantations into Golden Forests agroforestry projects represents a promising opportunity for diversification and growth. By adhering to national standards, optimising site selection, implementing best management practices, and leveraging regional development plans, the Company can establish a sustainable and profitable Durian operation. This plan serves as a foundational guide for Golden Forest's strategic planning and implementation efforts in the Durian sector.

APPENDIX K

The ten-year life cycle of *Aquilaria crassna* Agarwood

Image 1
Aquliaria crassna trees will commence fruiting after 7 years. These fruits normally would have two seeds. When the fruit splits, the seeds will be scattered by the wind, hence the difficulty in their collection.

Seeds can be stored in a refrigerator for two months. (Normally, seeds start to lose their germination capacity within one week. Germination rate 80~90% within one week. This indicates that the seed storage of *Aquilaria* is very difficult and should be sourced from very reliable suppliers. The average number of seeds from one kilo ranges between 3000 and 5000.

Image 2
The tail of the seeds faces upwards. The seeds may either be germinated in seedballs or individually germinated in trays with about 5mm deep in cocopeat (option is a mixture of soil, sand and manure) and kept under light shade. Germination starts after 10-12 days and is normally complete after one month.

Image 3
After 3-4 weeks of germination, the germinated seedlings are re-potted into polybags consisting of vermicompost, topsoil, rice husks and topsoil. At this age, it is considered the optimum time to export as the roots are not yet fully established, and subsequently the shock of being shipped bare-rooted and then re-potted is of minimal stress to the seedling, and results in a low mortality rate (depending on transit times). Germinated seedlings held in the nursery will be held in tunnels to maximise humidity and protection from heavy rains, before being exposed to shaded sunlight in the nursery.

Image 4
At the age of three months, the seedlings produce multiple leaves that allow the uptake of foliar nutrients and photosynthesis from the sunlight. Once the seedling reaches six months old, the stems are well hardened and brown, and the rooting structure is well established. It is recommended to repot the seedling into 6 x 6 x 10 inches to allow root development and prevent the roots from coiling.

Image 5
The ideal time to outplant agarwood seedlings is when it has reached a height of between 65 and 90cm. Normally, this is eight months from the date of germination. The recommended spacing is 3m x 2m (1667 per hectare). This allows for Cassava to be planted between the saplings to create shade during the first two years and then removed in year three. Alternatively, the agarwood saplings can be planted in between plantations, providing there is sufficient access to sunlight.

Image 6
The agarwood trees require a high level of maintenance during the initial three years. In addition to shade cover (year 1 and 2), there needs to be regular ring weeding, branch pruning, fertilisation, drip irrigation and disease management. By year five, the trees will reach a height of between 7-10 metres and a diameter of 10-15cm at breast height (dbh), the recommended size for inoculation to be applied.

Image 7
Agarwood inoculation involves deliberately inducing a tree's defence response to create the valuable agarwood resin by introducing a specific type of infection, like fungal or bacterial, through wounds or inoculation. The most common technique is dispersing wounding by mechanical injury: to drill several tens (20-100) of holes into which the inoculant is poured. In this case, induced agarwood is formed around these spot injury sites. The resulting agarwood from the localised formation is usually in the form of thicker wood chips. Inoculation through injecting fungi into the xylem vessel of the tree is another technique, however this has been known to cause wood rot.

Image 8
The agarwood tree is cut into blocks, and then the white wood is shaved away to access the resin/oud that has been formed from the inoculation process. It is recommended to harvest 18 to 24 months after inoculation to produce higher yields of resin/oud. The wood chips with a high concentration are primarily sold for the scent produced once they are burnt. Low-grade wood chips are used to produce agarwood essential oil. Approximately 140 kilos of wood chips will produce one litre of essential oil.

APPENDIX L

Frequently asked questions

Global demand for mangoes

Mangoes represent one of the world's most consumed fruits, with global annual consumption exceeding 45 million tonnes. Demand for premium varieties is increasing substantially in developed markets with limited domestic cultivation, including Japan, South Korea, the United States, and Europe.

The Philippines is strategically positioned as a major supplier of premium Carabao Mangoes, recognised internationally for exceptional sweetness (Brix levels of 18-24°). Export markets continue to develop, supported by improved logistics, streamlined phytosanitary protocols, and growing consumer preference for year-round premium tropical fruit availability.

Global demand for agarwood and oud essential oil

Agarwood ranks among the most valuable natural commodities globally, with the international market valued at approximately USD 41 billion and projected to reach USD 69 billion by 2029. Demand spans luxury fragrance, traditional medicine, religious ceremonies, and high-end decorative items.

Major luxury fragrance houses, including Tom Ford, Chanel, Dior, and Gucci, incorporate oud in premium product lines, with over 1,000 new oud-based fragrances launched annually. The Middle East represents the largest established market, whilst growing demand in China, Japan, and Southeast Asia, combined with limited natural supply and CITES regulatory restrictions, supports sustained premium pricing for cultivated agarwood from verified sustainable sources.

Overview of Golden Forests

Golden Forests is an agroforestry investment platform delivering professionally managed, high-value crop plantations across Asia with projected annualised returns of 18-20%.

2026 Focus:

- Agarwood and Carabao Mango cultivation in Southern Zambales, Philippines
- Durian cultivation launching 2027

Platform Approach:

Our Agroforestry Intelligence operational framework is designed for multi-crop, multi-region scalability, enabling expansion into additional premium agroforestry species and geographies as market opportunities emerge.

Key Differentiators:

- Exclusive licensing for advanced cultivation technologies
- Proprietary genetics and elite variety development
- AI-driven plantation management platform

- First-mover regulatory advantages
- Structured reforestation programme (1:1 native tree planting)

Vision:

Lead sustainable agroforestry in Asia through Agroforestry Intelligence, combining artificial intelligence, advanced science, and elite genetics to create long-term value, resilient ecosystems, and thriving communities.

Mission:

Position sustainable agroforestry as a compelling alternative investment, delivering outstanding long-term returns whilst creating local impact and environmental sustainability.

Transform Asian tree plantations into internationally recognised, high-value assets through sustainable practices, world-class production, and community partnership.

Core Values:

Integrity: Transparency and ethical conduct
Operational discipline: Structured management ensuring performance
Responsible stewardship: Sustainable land use and environmental protection
Quality and precision: Scientific cultivation meeting international standards
Long-term value creation: Sustainable performance aligned with responsible investment
Community development: Employment and social initiatives supporting local resilience

Our objectives

Financial Performance: Generate sustainable returns through advanced agroforestry science and AI-enabled management
Regulatory Excellence: Maintain full compliance with robust governance across all jurisdictions
Market Growth: Build a geographically diversified client base through disciplined expansion
Operational Innovation: Develop scalable plantation models integrating sustainability and yield optimisation
Stakeholder Confidence: Uphold rigorous governance, financial controls, and transparent reporting

Commitment to sustainability and ethical practices

Sustainability is central to Golden Forests' operations. The Company's structured reforestation programme plants one native endemic species (Narra, Agoho, Mangono, Molave, Talisay, Belite) for every agarwood and mango tree acquired by clients, supporting ecosystem restoration and biodiversity enhancement in Southern Zambales.

Research and innovation goals

Golden Forests maintains collaborative relationships with leading Philippine agricultural research institutions, including President Ramon Magsaysay State University (PRMSU), Visayas State University (VSU), and University of the Philippines Los Baños (UPLB), supporting advanced cultivation research, elite variety development, and sustainable agroforestry innovation.

Ownership and management of trees

Investors retain ownership of trees purchased, with Golden Forests managing them on their behalf.

- Ownership remains protected even if the company ceases operations, with a new management company appointed if necessary.
- Investors can transfer ownership or inherit trees as part of their estate.

Financial strength and investment model

Golden Forests operates debt-free, ensuring stability and control over its plantations.

- The company benefits from strategic support from experienced investors, reinforcing confidence in its financial management.
- Selling trees to investors allows for shared profits while maintaining full ownership and control.

Harvest management and returns

Clients receive net harvest proceeds (after 10% commission on agarwood oud wood chips, 20% commission on mango fruit) in accordance with the Sales and Management Contract. Agarwood harvest proceeds are distributed following years nine and ten (50% of portfolio each year). Mango harvest proceeds are distributed annually from year five through year twenty-five.

Advantages of investing in agroforestry

Agarwood and Carabao Mango cultivation in the Philippines benefits from favourable agro-climatic conditions, including a tropical climate with appropriate rainfall distribution (1,500-2,500mm annually), well-drained volcanic soils, competitive agricultural labour costs, and government support for agricultural development. Golden Forests' plantations deliver projected annualised returns of 18% (agarwood) and 12-20% (mango), comparing favourably to historical stock market returns (~10%) whilst providing tangible, biological asset backing.

Transparency and investor engagement

Golden Forests offers a structured client plantation visit programme in Southern Zambales, Philippines. The Company hosts a two-day site programme including guided plantation tours, operational briefings, and discussions with management and technical teams. Clients have access to GPS-verified tree locations, annual reporting, and secure client portal access supporting operational transparency and informed participation.

Realistic expectations for returns

Golden Forests' plantation investments offer projected annualised returns of 18% (agarwood, 10-year cycle) and 12-20% (mango, 25-year cycle), backed by tangible biological assets and professionally managed operations. Returns are derived from actual harvest sales of premium oud wood chips and Carabao Mango fruit to established international markets, providing an alternative to traditional financial assets with comparable or superior return profiles combined with measurable environmental impact.